Evaluating Data Centers as Local Energy System Assets: A System-Integrated PUE Approach

Sambeet Mishra[a], Ankana Singha[a], Khaled Aleikish[a], Thomas Øyvang[a], Dominic Stephen[b], Rob Hovsapian[c]

[a] *University of South-Eastern Norway, Porsgrunn, Norway*
[b] *Bax Innovation, Barcelona, Spain*
[c] *Savannah River National Laboratory, USA*

Abstract

Data centers are becoming material actors in local electricity and heat systems. They are usually reported as large electricity consumers, but the same sites can also export waste heat, provide flexible load, and use storage to reduce stress on local networks. The difficulty is that the dominant operational metric, Power Usage Effectiveness (PUE), treats all non-IT electricity as overhead. A data center that installs heat pumps, heat exchangers, controllability, or storage can therefore look less efficient under PUE even when the design creates useful external value.

This paper has two parts. Part I synthesizes policy, market, and project evidence on how data-center energy characteristics shape local business impacts in Europe. Part II defines and tests the central contribution: *A PUE-compatible companion metric that converts independently valued local-system benefits, including heat, flexibility, peak, carbon damage, and delay, into an energy-equivalent numerator credit.* We call this metric System-Integrated PUE (siPUE). The metric is tested in a two-stage design and operation framework with scenario-perfect benchmarks and a shared normalized min–max regret design. The min–max regret model is not proposed as a new optimization algorithm; it is used as a disciplined uncertainty testbed for evaluating metric behavior. The model is evaluated for four archetypes and ten shared uncertainty states using annual representative-day profiles, a stylized three-node DC-OPF electricity network, heat-recovery constraints, lifecycle delay accounting, and a lease or opportunity-cost representation of land.

The regenerated evidence bundle supports a conservative interpretation. In the no-services case, siPUE reduces exactly to PUE. When integration

services are active, PUE and siPUE can move in opposite directions: the robust design raises average PUE relative to the no-services case, while average siPUE remains close to the no-services value because heat, flexibility, and avoided-system-cost credits offset part of the additional facility energy. A compact metric benchmark compares PUE, delay-adjusted siPUE, post-construction siPUE, and system benefit per MWh IT; carbon, heat, and emissions are retained as separate ledgers rather than re-labelled as standard KPIs. Train/test and out-of-sample diagnostics show feasible held-out operation and nonnegative comparable regret, but these checks are treated as internal reproducibility evidence rather than external validation. The results argue for reporting PUE and siPUE together: PUE remains a useful internal-efficiency measure, while siPUE reveals system value that PUE hides in service-active designs.

Keywords: Data centers, Power Usage Effectiveness, System-Integrated PUE, waste-heat reuse, flexibility, robust optimization, local energy systems

---

Variables	
GRID_t	Grid electricity import at time t (MWh)
IT_t	IT electricity served at time t (MWh)
HEAT_t	Heat exported to district heating at time t (MWh_{th})
FLEX_t	Flexibility service provision at time t (MW or MWh equivalent)
$\mathrm{BCH}_t,\ \mathrm{BDIS}_t$	Battery charge and discharge at time t (MWh)
SOC_t	Battery state of charge at time t (MWh)
G_n^t	Thermal generation dispatched at network node n at time t (MWh)
F_ℓ^t	Power flow on interconnector ℓ at time t (MWh)
K_{HX}	Installed heat exchanger capacity (MW_{th})
K_{FLEX}	Installed flexibility capacity (MW)
K_{BAT}^E	Battery energy capacity (MWh)
K_{BAT}^P	Battery power capacity (MW)
η	Efficiency improvement factor (fractional reduction in overhead)

Parameters

$p_t^{\mathbf{IT}}$	IT service revenue per delivered IT energy at time t (EUR/MWh)
$p_t^{\mathbf{el}}$	Electricity price at time t (EUR/MWh)
$p_t^{\mathbf{heat}}$	Heat price at time t (EUR/MWh$_{th}$)
$p_t^{\mathbf{flex}}$	Flexibility price at time t (EUR/MW or EUR/MWh)
$p_t^{\mathbf{CO2}}$	Carbon price at time t (EUR/tCO$_2$)
$c_t^{\mathbf{fuel}}$	Fuel cost for thermal generation at time t (EUR/MWh$_{th}$)
$c_n^{\mathbf{vom}}$	Variable O&M cost for thermal generator at node n (EUR/MWh)
$c^{\mathbf{trans}}$	Transmission usage cost per unit of power flow (EUR/MWh)
γ_t	Grid carbon intensity at time t (kgCO$_2$/MWh)
ε_n	Emission factor for thermal generator at node n (tCO$_2$/MWh$_{th}$)
η_n	Thermal efficiency of generator at node n (dimensionless)
$\mathbf{IT}_t^{\mathbf{base}}$	Baseline IT demand at time t (MWh)
$\mathbf{DH}_t^{\mathbf{cap}}$	District heating export capacity at time t (MWh$_{th}$)
$\mathbf{GRIDCAP}_t$	Grid import capacity at time t (MW)
ϕ	Recoverable heat fraction (dimensionless)
$\mathbf{COP_{HP}}$	Heat pump coefficient of performance (dimensionless)
$\eta_c,\ \eta_d$	Battery charge and discharge efficiency (dimensionless)
w_t	Representative-day weight for hour t (dimensionless)
σ	Construction-delay scale factor: $\sigma = (1-\Delta_{\text{cfg}})(1-\Delta_{\text{sc}})$ (dimensionless)
$\Delta_{\mathbf{cfg}}$	Global construction-delay fraction (dimensionless)
$\Delta_{\mathbf{sc}}$	Scenario-specific construction-delay fraction (dimensionless)

$p_{\mathbf{ref}}$	Frozen no-services electricity-resource reference price used to convert system benefit into the siPUE energy credit (EUR/MWh)
$c_{\mathbf{land}}$	Annual land cost per unit area (EUR/m^2/yr)
$\phi_{\mathbf{maint}}$	Land maintenance markup (dimensionless)
L_0	Baseline land footprint of the data center (m^2)
$a_{\mathbf{HX}}, a_{\mathbf{FLEX}}, a_{\mathbf{BAT}}$	Land footprint coefficients per unit capacity (m^2/MW or m^2/MWh)
$c_{\mathbf{demand}}$	Demand charge rate (EUR/MW/yr)
λ	Regret–profit scalarization weight (dimensionless)

Aggregate cost and revenue terms

$C_{\mathbf{demand}}$	Annual peak-demand charge (EUR)
$C_{\mathbf{capex}}$	Annualized capital expenditure for integration assets (EUR)
$C_{\mathbf{eff,capex}}$	Annualized capital cost of efficiency improvements (EUR)
$C_{\mathbf{eff,benefit}}$	Operational savings from efficiency improvements (EUR)
$C_{\mathbf{land}}$	Annual land cost (EUR)
$C_{\mathbf{penalty}}$	SLA shortfall penalty for unserved IT load (EUR)
$E_{\mathbf{FAC}}$	Total facility electricity consumption (MWh)
$E_{\mathbf{IT}}$	Total IT electricity consumption (MWh)
$E_{\mathbf{credit}}$	Energy-equivalent credit used in siPUE (MWh)
$C_{\mathbf{heat}}$	Total heat export revenue (EUR)
$C_{\mathbf{flex}}$	Total flexibility service revenue (EUR)
$C_{\mathbf{CO2}}$	Total CO_2 cost from grid carbon exposure (EUR)
Π	Total annual profit (EUR)
$\Pi_s^\star$	Perfect-foresight optimal profit for scenario s (EUR)
Z	Worst-case regret across scenarios (EUR)
$\bar{\Pi}$	Average profit across scenarios (EUR)

1. Introduction

Electricity demand from European data centers has become a material planning variable for both transmission and distribution networks. Commission estimates put EU consumption at 76.8 TWh in 2018, with projections of 98.5 TWh by 2030 under a trend scenario and up to 160 TWh under a high-growth case [1]. More recent assessments suggest consumption already reached roughly 96 TWh in 2024 and could approach 168 TWh by 2030 [2].

Growth is reinforced by traffic expansion and a shift toward compute-intensive workloads. Artificial intelligence (AI) workloads are expected to accelerate demand, with energy needs rising rapidly during 2023–2030 [3, 4]. If these trajectories persist, the data-center share of EU electricity could increase from about 3.1% to around 5.7% by 2035 [2], making aggregate demand comparable to national consumption in several member states [5, 6]. At that scale, the build-out becomes a business and investment problem under binding local constraints: connection rights and congestion on the power side, offtake and tariff interfaces on the heat side, eligibility rules for flexibility markets, and land-use approvals, permitting timelines, and community acceptance at the site level increasingly shape which projects can proceed and how quickly capital can be deployed [7]. Integration features can partially mitigate these frictions by delivering local co-benefits (e.g., heat export) or by reducing peak stress through controllability and storage, but they require additional interfaces and investment that must be evaluated on a comparable basis.

In parallel, the EU energy transition is pushing electricity and heat systems toward higher renewable shares, electrification, and sector coupling [8]. Policy packages emphasize digitalization and flexibility as enabling layers for this transition [9], while sustainability reporting frameworks for large data centers are being formalized through Directive (EU) 2023/1791 and industry initiatives [10, 11]. This setting creates a dual role for data centers: treated as inflexible loads they can exacerbate congestion, but when designed for integration they can supply valuable services such as waste-heat export and grid support [12, 13, 14, 15].

Data-center participation in heat and flexibility markets is governed by tariffs, market access rules, and contract structures. Yet the dominant reporting metric, PUE, evaluates only internal electricity overhead and does not account for externally valued services. As a result, integration investments that increase electricity use to deliver heat or flexibility can worsen

reported efficiency even when they improve economic and system outcomes. This creates a reporting–investment tension that motivates the development of a PUE-compatible companion metric capable of translating independently valued local-system benefits into the same energy-normalized language as PUE.

AI-driven data centers add another layer to this problem. Training, inference, and data-transfer workloads can have different electrical and thermal signatures, and operators increasingly need controls that respond to fast changes in power demand, cooling demand, and service availability. PUE alone cannot say whether the additional electricity used by such controls is wasteful overhead or the cost of delivering useful flexibility, heat recovery, or reliability services. That distinction is central to the metric problem studied in this paper.

1.1. Contributions

This paper makes four contributions:

1. It integrates evidence on data centers and local energy systems, emphasizing the mechanisms through which energy characteristics shape local business impacts and investment readiness.

2. It defines and tests System-Integrated Power Usage Effectiveness (siPUE): a PUE-compatible companion metric that converts independently valued local-system benefits, including heat, flexibility, peak, carbon damage, and delay, into an energy-equivalent numerator credit.

3. It uses a two-stage optimization model with scenario-perfect benchmarks and a shared normalized min–max regret design as the uncertainty testbed for evaluating the metric, with lifecycle delay, carbon ledgers, land accounting, heat recovery, flexibility, storage, and DC-OPF network constraints.

4. It evaluates siPUE for four archetypes and ten shared uncertainty states, reports a compact PUE–siPUE metric benchmark, decomposes the siPUE credit by ledger component, and tests sensitivity to the reference price, heat price, flexibility price, cooling overhead, land availability, and construction delay.

1.2. Related Work

The literature relevant to this work spans four major areas: (i) data centre efficiency metrics, (ii) waste-heat recovery and district-heating integration, (iii) data centre flexibility and demand response, and (iv) multi-energy integrated systems. These streams are reviewed chronologically and highlight the metric gap that motivates System-Integrated PUE (siPUE).

1.2.1. Foundation of Data Centre Metrics (2008–2012)

The first generation of data centre metrics was established through The Green Grid between 2008 and 2012, culminating in the widely adopted Power Usage Effectiveness (PUE) metric [16, 17]. PUE provided a simple ratio of total facility energy to IT energy and played a crucial role in raising awareness about infrastructure efficiency. However, its design implicitly assumed that (i) all non-IT energy is undesirable, (ii) energy source differences are irrelevant, and (iii) time-varying system behaviour does not matter. These assumptions became problematic as data centres began interacting with electricity markets, district heating networks, and flexibility services.

1.2.2. First Critical Analyses and Metric Extensions (2015–2017)

By 2015–2017, several works began revealing structural weaknesses in PUE. Aravanis and Voulkidis [18] highlighted the absence of flexibility considerations. Reddy et al. [19] provided a comprehensive taxonomy of sustainability metrics and demonstrated that PUE captures only one dimension of data centre performance. Van de Voort et al. [20] showed empirically that PUE varies with cooling architecture and climate, undermining fair cross-facility comparisons.

During this period, additional single-dimension metrics emerged: Carbon Usage Effectiveness (CUE), Water Usage Effectiveness (WUE), and Energy Reuse Factor / Energy Reuse Effectiveness (ERF/ERE). These standards (ISO 30134 series) addressed specific dimensions but remained fundamentally parallel indicators, without a unified or market-based way to aggregate them.

1.2.3. Waste-Heat Recovery and District Heating Integration (2018–2020)

Research into waste-heat utilisation expanded significantly as data centres grew larger and more thermally intensive. Early foundational work by Brunschwiler et al. [21] demonstrated the technical viability of high-grade heat recovery using direct liquid cooling. Wahlroos et al. [12] analysed the

integration of data centre waste heat into Northern European district heating (DH) systems, identifying substantial potential for system-wide carbon reductions. Copertaro et al. [22] positioned data centres as prosumers that both consume electricity and export heat. These works establish heat reuse as a genuine system service but do not provide an integrated metric capable of valuing it alongside energy, carbon, or flexibility. The literature still lacks a reporting-compatible way to translate heat export into an equivalent, market-aligned sustainability score.

1.2.4. Data Centre Flexibility and Demand Response (2019–2021)

With increasing renewable penetration, data centres became candidates for providing grid flexibility. Liu et al. [23] demonstrated how IT load modulation (e.g., CPU scaling) can deliver demand response. Antal et al. [24] proposed an explicit flexibility management framework for data centres.

However, these works analyse flexibility in isolation, without linking it to heat reuse, carbon intensity, or facility-wide sustainability. Furthermore, they often rely on engineering assumptions regarding flexibility prices rather than settled service-specific market values. This gap becomes evident in Crozier and Liska [25], who find that real-world data centres exhibit weak price responsiveness, highlighting the need for econometric estimation of flexibility value.

1.2.5. Integrated Energy Systems and Sector Coupling (2020–2024)

Research shifted toward integrated multi-energy systems around 2020. Zhang et al. [26] surveyed the integration of data centres with renewable energy. Jerez Monsalves et al. [27] showed that flexible cooling and waste-heat recovery from data centres reduce system costs and emissions in the Danish energy system. Wang et al. [28] presented the "Data Centre Integrated Energy System" (DC-IES) concept, combining electricity, heat, cooling, renewables, and storage.

These studies quantify system-level impacts but do not produce metrics. Their economic valuation is embedded in optimisation models rather than expressed as standalone, operator-focused indicators.

1.2.6. Holistic Sustainability Metrics (2023–2025)

Attempts to provide broader sustainability scoring emerged more recently. Terenius [29] introduced the Datacenter Energy Sustainability Score (DESS), which incorporates waste heat, energy sources and social context. However,

DESS is qualitative, not market-based, and cannot be used for hourly operational decisions. Newer reviews of waste heat [30, 31] and data centre flexibility [32] reinforce the need for quantifiable, energy-system-oriented indicators.

1.3. Synthesis and Gap Identification

Table 1 positions the proposed metric against the closest reporting metrics and integrated-system modelling literature. The novelty is not another efficiency ratio, and it is not the min–max regret optimizer. The novelty is the accounting translation from local-system value into a PUE numerator credit.

Across this literature, three persistent gaps remain:

- Fragmentation: Existing metrics (PUE, CUE, WUE, ERF/ERE, REF, DESS, DCRE) each measure one dimension or resource bundle but do not provide a PUE-compatible companion score for local-system services
- Lack of market valuation: Heat reuse, flexibility, and carbon impacts are rarely valued using empirical electricity, heating, reserve or ETS price data; most studies rely on engineering assumptions or ad hoc pricing
- Absence of a PUE-compatible local-system credit: Existing work does not provide a transparent numerator credit that converts independently valued heat, flexibility, peak, carbon-damage, and delay effects into the same energy-normalized language as PUE

1.4. Conceptual Framework and Research Questions

The paper is structured around a conceptual framework that links energy characteristics, data center design decisions, and local energy system outcomes. Evidence from policy documents, market data, and industry practice provides qualitative and institutional context for the energy characteristics and business readiness layer, while the model formalizes the data center responses and quantifies system outcomes. This leads to the following research questions:

1. RQ1: How do energy characteristics and policy frameworks shape local business impacts of data centers in Europe

Table 1: Comparison of siPUE with adjacent data-centre metrics and modelling approaches.

Approach	Primary function	Values local-system services?	PUE numerator credit?	Gap relative to siPUE
PUE [16, 17, 33]	Facility electricity divided by IT electricity	No	Baseline ratio, not a credit	Treats heat-pump, storage, and control electricity as overhead even when they create external value.
ERF/ERE [33]	Reused-energy reporting and energy-reuse effectiveness	Heat only, in energy terms	Partial reuse adjustment	Does not value flexibility, peak, network, carbon-damage, or delay effects.
CUE [19]	Carbon emissions per IT energy	Carbon footprint only	No	Parallel carbon-intensity indicator rather than a PUE-compatible local-system credit.
WUE [33]	Water consumption per IT energy	No	No	Tracks water use outside the energy-system value boundary.
REF [33]	Renewable-energy share	No	No	Tracks procurement mix rather than local heat, flexibility, peak, carbon-damage, or delay value.
DESS [29]	Broader sustainability score	Partial, contextual	No	Not a market-valued companion metric and not directly reducible to PUE.
DCRE [34]	Resource-effectiveness view of energy, water, water stress, and climate context	Indirect	No	Broader resource metric, not a local-system service credit for PUE reporting.
Integrated energy-system optimisation papers [27, 28]	System-cost or emissions optimisation	Yes, inside the model	No	Quantifies system impacts but does not define a standalone reporting-compatible numerator credit.
siPUE (this study)	Companion metric for service-active data centres	Yes: heat, flexibility, peak, carbon damage, and delay	Yes	Converts independently valued local-system benefit into an energy-equivalent numerator credit and reduces to PUE in the no-services case.

2. RQ2: Can a PUE-compatible companion metric reveal local-system value from heat, flexibility, peak, carbon damage, and delay while preserving the no-services PUE identity

3. RQ3: How do robust, profit-seeking design decisions shift traditional efficiency metrics and system-integrated metrics

The remainder of the paper addresses RQ1 through a structured synthesis of policy, market, and project evidence (Part I) and RQ2–RQ3 through the metric definition and model-based analysis (Part II).

2. Part I: Data Centers in Local Energy Systems — Policy and Market Context

This section examines how data centers interact with local energy systems from a policy and business-investment perspective. Data-center electricity use has become a material planning variable in Europe. Commission assessments place EU consumption at 76.8 TWh in 2018, with projections of 98.5 TWh by 2030 under a trend case and up to 160 TWh under a high-growth case [1]. Industry reporting suggests that demand already reached roughly 96 TWh in 2024 and could approach 168 TWh by 2030 [2]. These trajectories are driven by sustained cloud growth, accelerating data traffic, and the energy intensity of AI workloads [3, 4, 35]. At the facility level, non-IT energy remains significant; cooling and auxiliary systems typically account for roughly 30–40% of total electricity consumption [36]. The combination of scale and growth means that data-center siting and design decisions now shape local grid planning and regional investment pipelines.

At the same time, Europe is pursuing decarbonization, electrification, and sector coupling, which require more flexible and digitally coordinated energy systems [8, 9, 37]. Data centers are not only large loads but also as potential local system assets that can provide waste-heat recovery, flexible demand, and grid services when integration pathways exist. This duality creates a business challenge: integration can unlock new revenues and de-risk grid access, but it also adds capital costs and exposes projects to policy and market uncertainty.

Part I therefore adopts an explicitly business-oriented lens. The macro–meso–micro structuring adopted in this study is informed by the multi-level perspective (MLP) on socio-technical transitions [38]. Projects must reach financial close under uncertainty about connection rights, tariff treatment,

reporting requirements, and partner commitments. The multi-level governance framework is used to trace how data-centre innovations can scale into broader socio-technical transitions. Micro-level projects generate learning and proof of concept, meso-level actors (e.g., utilities and regional authorities) enable replication through rules and delivery models, and macro-level policy and funding shape long-term direction. This lens fits the data-centre context because scaling depends not only on technology, but also on enabling policies, workable organisational models, and access to low-cost, standardised finance. The analysis is organized around (i) binding constraints that shape site selection and time-to-market, (ii) the service channels that enable value stacking, and (iii) the cross-level governance conditions that determine whether integration becomes bankable. This framing motivates why a reporting-compatible metric that accounts for externally valued services is needed for coherent investment and performance assessment.

The investment case for system-integrated data centers is shaped by interlocking constraints that operate at different governance levels. Macro policy establishes regulatory certainty and carbon-price trajectories; meso infrastructure determines physical feasibility and sequencing; micro finance translates the resulting risk profile into capital costs and revenue requirements. These layers are not independent. Policy can create incentives that lower financing costs, but only if local infrastructure exists to deliver services and project-level contracts can lock in revenue.

2.1. Macro level — Policy and market architecture

The macro level refers to EU and national policy priorities, regulation, and funding frameworks that shape incentives for data-centre energy integration. Stakeholders include EU institutions, national ministries, regulators, and public funding bodies. At the macro level, EU climate policy provides the overarching regulatory frame for integration. The European Green Deal and the "Fit for 55" package establish binding decarbonization targets that indirectly increase the value of electricity efficiency, waste-heat reuse, and flexibility services [39, 8]. The Energy Efficiency Directive recast (Directive (EU) 2023/1791) introduces explicit reporting obligations for large data centers and codifies sustainability KPIs such as PUE, Renewable Energy Factor, and Energy Reuse Factor [10, 33]. Voluntary commitments such as the Climate Neutral Data Centre Pact reinforce these objectives and translate them into sector-level targets for renewable supply and heat reuse [11]. These policy instruments matter not only as compliance obligations but as governance

signals: reporting metrics become part of lender and investor due diligence, shaping reputational risk, access to capital, and eligibility for public support.

Standardized KPI disclosure also functions as a capital signal. Reporting comparability affects ESG scoring, due diligence, and financing conditions. Where reporting frameworks emphasize internal efficiency without recognizing system services, integration investments may appear less performant despite improving system outcomes[10, 11, 33, 17]. This misalignment between public objectives and private performance metrics can weaken incentives for system-integrated design.

Digitalization policy is the second macro pillar. The EU action plan for digitalizing the energy system frames flexibility, interoperability, and data availability as prerequisites for integrating variable renewables and coordinating distributed assets [9]. This positions data centers as potential "digital-energy hubs" whose operational telemetry, controllability, and digital interfaces are integral to system services. Macro-level policy thus creates an incentive landscape in which data centers that can provide measurable services are more likely to qualify for preferential treatment or supportive tariffs, while those that remain inflexible loads face higher regulatory risk or stricter connection conditions.

Policy-induced investment incentives sit alongside these reporting obligations. Regional initiatives and public–private programs aim to accelerate investment readiness by providing blueprints, de-risking mechanisms, or bundled project pipelines [40, 41]. The business impact of such programs depends on their stability and on whether they can shorten the time to contract and commissioning. Where incentives are credible and durable, they reduce financing risk; where they are uncertain, they can amplify the option value of waiting.

Carbon pricing adds a third macro channel. The EU ETS has measurable impacts on electricity-system dispatch and price formation, with carbon price trajectories shaping the relative value of demand reduction, heat reuse, and electrified heat pumps [42]. For data centers, carbon exposure becomes a regulatory risk that influences the expected marginal cost of electricity and the value of low-carbon energy supply. This policy risk is not purely environmental: it affects the discount rates applied to future cash flows and the choice between integration investments and conventional efficiency upgrades. A stable carbon-price trajectory can support long-term contracts for heat and flexibility; a volatile trajectory increases revenue uncertainty and can delay investment.

Regulatory certainty compounds across these macro channels: because data centers are long-lived assets, durable policy signals reduce financing risk while volatile signals raise it[8, 10, 11], a dynamic developed in financial detail at the micro level below.

Finally, electricity market design determines whether flexibility can be monetized and whether congestion relief is rewarded. Studies on data-center flexibility show that market access and product design are decisive for realizing value from demand response and ancillary services [43, 13, 32, 25]. Where balancing markets are open and telemetry requirements are feasible, data centers can monetize battery and load flexibility. Where participation rules are opaque or markets are illiquid, flexibility remains a theoretical potential rather than a bankable revenue stream. Macro policy thus sets the boundary conditions for value stacking, but it does not guarantee investable outcomes.

2.2. Meso level — Regional infrastructure and spatial constraint

The meso level covers regional and local governance and market arrangements that turn policy into planning, tariffs, and investable projects. Stakeholders include regional/metropolitan authorities, DSOs/TSOs, district-heating operators, aggregators, and local permitting bodies. Meso level conditions determine whether macro incentives can be converted into projects. The binding constraint is often grid capacity rather than annual energy. Large, near-constant loads can trigger network reinforcements, and in some regions grid operators have constrained or paused new data-center connections due to local network limits [7]. Congestion pricing and connection-rights frameworks influence both operating costs and the feasibility of additional load, making grid access a strategic asset rather than a commodity. Regional transmission and distribution operators increasingly treat large data centers as sources of local stress, which means that permitting and connection rights depend on credible mitigation strategies and co-benefits [6, 7]. This logic directly affects capital allocation: projects with grid-relief features (flexibility, behind-the-meter storage, or verifiable heat export) are more likely to clear permitting and avoid costly grid upgrades.

Congestion is not only a planning issue; it is a pricing and contract issue. Where congestion charges or locational tariffs are applied, the effective electricity cost can vary materially across sites and over time, and grid import caps can become binding constraints that limit expansion options. These locational signals influence both revenue projections and risk assessments,

especially for projects that rely on tight operating margins. For developers, the implication is that congestion relief and flexibility can carry real option value by protecting against future price volatility and curtailment risk [6, 7].

Permitting and local acceptance further tie congestion and land constraints to business outcomes. Municipalities increasingly evaluate data-center projects through the lens of local co-benefits, including heat reuse, grid services, or renewable integration. Projects that can credibly demonstrate such co-benefits often face lower permitting friction and stronger stakeholder support. This turns integration into a strategic permitting asset rather than a purely operational choice.

Land is a parallel constraint. Large facilities require substantial plots, and integration equipment—heat pumps, heat exchangers, thermal storage, battery containers, and electrical substations—expands the footprint and the need for utility corridors. Land availability is not only a physical constraint but a capital constraint: siting in land-scarce zones raises acquisition costs, extends permitting timelines, and forces trade-offs between integration depth and real-estate feasibility. Because location choices are sticky, land constraints create spatial lock-in: once a site is committed, the ability to exploit local heat networks, renewable resources, or grid capacity is fixed for the project's life. This lock-in is especially relevant for projects that require proximity to district heating networks or industrial offtakers to monetize waste heat [14, 15].

Land constraints also shape expansion options. Integrating heat recovery, storage, or on-site substations often competes with future IT expansion or with other land-intensive infrastructure such as logistics or water-treatment systems. In urban or industrial clusters where land prices and zoning restrictions are tight, these trade-offs can shift investment decisions toward smaller integration packages or phased deployments. The business implication is that land is not merely a siting input but a capacity-limiting factor that affects both near-term financing and long-term optionality.

District heating infrastructure is a second meso constraint. Heat reuse is technically feasible in many contexts, but commercial viability depends on network proximity, temperature requirements, and utility tariff structures [22, 12, 14, 15]. In regions with established district-heating governance, standardized interconnection rules and repeatable contract templates reduce transaction costs and improve bankability [12, 15]. In regions without such infrastructure, heat recovery becomes a long-lead investment requiring coordination across municipalities, utilities, and private developers. These dif-

ferences are not captured by a single efficiency metric but directly influence capital requirements and revenue timing.

Infrastructure sequencing is central to this coordination challenge. Heat recovery projects often require parallel investments in heat-pump capacity, pipework, and network reinforcement. When the heat network expands more slowly than data-center build-out, the heat revenue stream is delayed or curtailed, which weakens the business case and can strand integration assets. Successful projects typically resolve sequencing risks through early-stage agreements on expansion schedules and tariff treatment, often involving municipal utilities as coordinating agents. This sequencing constraint mirrors grid reinforcement delays and reinforces the need to price construction delay and infrastructure timing explicitly in investment analysis.

Renewable resource geography also shapes meso feasibility. Nordic contexts provide lower ambient temperatures and proximity to low-carbon generation, improving both PUE performance and the economics of heat cascading [44]. Conversely, high-demand urban clusters often combine strong heat demand with constrained grid capacity and scarce land, raising the value of integration while making it harder to realize. Emerging "energy hub" concepts co-locate data centers with renewables, storage, and other flexible loads, using the data center as anchor demand to unlock investment in local generation [45]. These spatial factors make site selection a strategic business decision rather than a purely technical optimization.

Construction delay is the critical meso-level risk that links infrastructure constraints to financial outcomes. Grid reinforcements, zoning approvals, and interconnection negotiations can delay energization even when a facility is otherwise ready, creating a timing asymmetry between developers and network operators. Because data centers are capital intensive, delays destroy value by shifting revenue streams without reducing fixed costs. The risk is amplified when integration assets are built ahead of guaranteed market access or offtake contracts, which can strand capital for multiple years. This makes infrastructure sequencing and partner coordination central to investment readiness.

Timing asymmetry is particularly acute when grid reinforcement timelines exceed data-center construction timelines. In many jurisdictions, data centers can be built in 18–24 months, while network upgrades or district-heating expansions require multi-year planning cycles. This mismatch creates a coordination risk: capital can be deployed faster than enabling infrastructure, and the resulting idle capacity erodes IRR and reduces appetite

for integration. The report highlights that successful projects often rely on early-stage alignment between developers, utilities, and municipalities to synchronize these timelines and to lock in connection rights before major capital is committed.

2.3. Micro level — Project finance and capital allocation

The micro level refers to site-level technologies and contracts implemented by individual data centres and their partners. Stakeholders include data-centre operators, technology providers, heat off-takers, flexibility/market participants, and on-site energy service companies. At the micro level, investors evaluate data centers through cash-flow logic: Net Present Value (NPV), Internal Rate of Return (IRR) thresholds, and the ability to service debt under downside scenarios. Integration investments increase upfront CAPEX (heat pumps, exchangers, storage, telemetry), but can diversify revenue by adding heat-offtake and flexibility payments to the core IT business. The report highlights that diversified business models are central to investment readiness, particularly when heat and flexibility revenues are structured as contracted cash flows rather than speculative market exposure. Evidence from European heat-reuse projects shows that commercial structures range from utility-paid offtake contracts to co-investment models, with long-term price agreements often required to reach bankability [15]. On the flexibility side, value capture depends on market access and the ability to meet telemetry and performance requirements [43, 13].

Uncertainty increases the cost of capital. If energy prices, carbon prices, or market-access rules are unstable, investors apply higher discount rates or require stronger contractual guarantees. This raises the hurdle for integration assets relative to conventional efficiency upgrades. Delay risk exacerbates this effect: construction delays lengthen the period before revenue accrues and can breach debt-service covenants. The option value of waiting can therefore dominate near-term integration decisions, especially when the regulatory trajectory or grid-reinforcement timetable is uncertain. These dynamics make financing conditions as important as technical feasibility.

Debt structure and covenant design translate these uncertainties into capital allocation decisions. Projects that can demonstrate long-term contracted heat offtake or stable flexibility revenues are more likely to access lower-cost debt and higher leverage, while projects with speculative service revenues face equity-heavy structures and higher required returns. Integration therefore creates a trade-off: it raises CAPEX but can reduce long-run risk if ser-

vice contracts are credible. This trade-off is central to investment appraisal and underpins the need for metrics that translate system-service value into signals that financiers can use.

Performance metrics influence financing costs. Reporting obligations under the Energy Efficiency Directive and voluntary sector commitments institutionalize PUE and related KPIs as governance tools [10, 11]. Because lenders and investors increasingly rely on standardized metrics in ESG scoring and due diligence, KPI performance can affect the cost and availability of capital. The challenge is that conventional PUE treats all non-IT electricity consumption as overhead, which can penalize projects that add integration assets to generate external system value [17, 16, 20]. This creates a micro-level financing risk: projects that improve system outcomes may appear worse on standard metrics, potentially raising financing costs despite stronger long-term value.

These financing dynamics incentivize staged integration strategies. Developers may prioritize assets that secure connection rights or heat offtake first, while deferring optional assets (e.g., additional storage or advanced controllability) until market access or tariff structures are clearer. Integration becomes bankable when uncertainty is reduced through contracts, regulatory clarity, or pilot-scale validation. The model in Part II reflects this reality by separating regimes with active services from those where integration remains latent.

These characteristics imply that data centers simultaneously represent infrastructure pressure and integration opportunity. The following section conceptualizes the primary service channels through which integration occurs.

2.4. Service Channels and Revenue Stacking

Data centers contribute to local energy systems through three principal channels: heat recovery, electrical flexibility, and coordinated renewable anchoring. These channels correspond to potential value streams that can be stacked to improve project economics, but each requires distinct infrastructure interfaces, market access conditions, and contractual arrangements. The business relevance is direct: when service revenues are contractible and measurable, they can offset integration CAPEX and reduce dependence on volatile electricity prices; when they are not, integration risks becoming a compliance cost rather than a value stream.

Financing structures tend to mirror these revenue profiles. Where long-term offtake or flexibility contracts are in place, projects can access cheaper debt and higher leverage; where revenues are uncertain, equity shares increase and hurdle rates rise. The business consequence is that integration decisions cannot be separated from contract design and counterparty risk. Bankable integration often requires early-stage negotiation with utilities and municipalities to ensure that service revenues are recognized in the financial model rather than treated as speculative upside.

Measurement and verification are an additional, often underestimated, business requirement. Heat export requires calibrated metering and agreed accounting for temperature lift and losses; flexibility participation requires real-time telemetry and compliance with market performance metrics [9, 13]. These systems impose capital and operational costs that need to be internalized in the financial model. The EU digitalization action plan explicitly positions interoperable data standards and monitoring infrastructure as enabling conditions for flexibility and sector coupling [9]. For data centers, this implies that integration is partly a data-governance investment: without robust measurement and reporting systems, service revenues cannot be settled or audited, and financial risk increases.

Data centers reject low-grade heat that can be recovered from cooling circuits and upgraded for productive use. In a common configuration, a secondary water loop extracts heat via an exchanger and provides a low-temperature source stream (typically in the 25–40°C range) that is lifted by a heat pump to district-heating supply levels or used by nearby process-heat offtakers.

Figure 1 illustrates the physical and contractual structure of this service channel. The upper layer represents the thermodynamic pathway from IT load to heat delivery. The lower layer highlights the institutional interface required for monetization: connection to district heating infrastructure and contractual alignment with utilities or local offtakers.

Urban waste heat has been identified as a material resource for decarbonizing building heating when effectively integrated [14]. The recast Energy Efficiency Directive introduces reporting and feasibility assessment obligations for large facilities [10]. However, economic viability depends on tariff structures, network access, and regulatory certainty rather than technical feasibility alone.

A large share of data-center waste heat is technically recoverable under suitable configurations, but that monetization pathways are uneven across re-

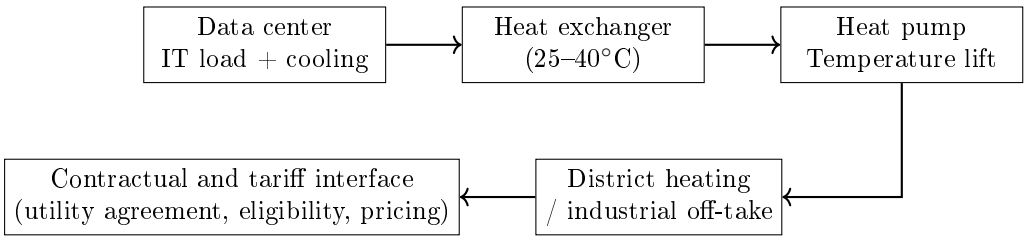


Figure 1: Process flow for recovery and upgrading of data centre waste heat for district heating or industrial off-take, including the contractual and tariff interface.

gions. Even where heat demand exists, mismatch between seasonal demand, temperature requirements, and network access can reduce achievable utilization factors. From a business perspective, the decisive variables are therefore not only heat availability but the ability to secure long-term offtake commitments and to allocate temperature-lift responsibilities across stakeholders (operator, utility, or third-party owner).

Financing structures reinforce this point. Heat-recovery assets often sit outside the data-center "fence line" and require co-investment with utilities or municipalities. In practice, bankable projects rely on long-term contracts that define pricing, curtailment rules, and performance obligations, which transforms heat export from an uncertain byproduct into a predictable revenue stream [15]. Where such contracts are absent, heat recovery is treated as discretionary, even if it improves system outcomes.

Project evidence indicates that heat recovery can be commercialized through multiple arrangements: utility-paid offtake into district heating, co-investment in heat pumps and network interfaces, or third-party ownership of the recovery assets with long-term heat purchase agreements. Best-practice surveys document operational and late-stage projects across diverse European contexts, suggesting that remaining barriers are often institutional (interconnection terms, tariff treatment, responsibility allocation) rather than purely technical [15]. In regions with mature district-heating governance, standardized interconnection rules and repeatable contract templates reduce transaction costs and improve investment readiness [12, 15].

Successful heat projects typically emerge where municipal utilities, developers, and offtakers coordinate early on tariffs and network expansion. Examples from Nordic and Central European cities show that heat reuse becomes commercially viable when district-heating operators can monetize lower-temperature supply and when data centers can internalize a share of

the avoided heat costs. These cases reinforce that heat recovery is not a technology-only decision; it is a multi-actor contracting problem that sits at the intersection of policy, infrastructure, and finance [15].

Heat pricing is often indexed to local fuel prices or to regulated district-heating tariffs, which means that the business value of heat export depends on broader energy-market conditions and carbon pricing. As carbon prices rise, the relative value of low-carbon heat increases, improving the economics of heat recovery even when electricity costs rise [42]. For data centers, this introduces a second price exposure beyond electricity: the heat price trajectory. This reinforces the importance of long-term offtake agreements that define price indexation and risk sharing between operators and utilities [14, 15].

Beyond thermal integration, data centers possess latent flexibility through controllable loads and battery systems originally installed for reliability. Under appropriate market rules, these assets can provide balancing and ancillary services [13]. Estimates suggest that European data centers could offer substantial demand response capacity if enabled by market design [43]. Demonstrated deployments illustrate that portions of battery capacity can participate in balancing markets without compromising uptime [35].

Flexibility participation depends on both technical capability and regulatory access. On the technical side, controllable cooling and workload shifting allow short-term reductions in demand without service disruption, while battery energy storage provides fast response for frequency regulation or reserve products [13, 24, 32]. On the regulatory side, eligibility criteria, telemetry requirements, and settlement intervals determine whether data centers can participate at scale [25]. Where these conditions align, flexibility can become a stable revenue stream that also reduces congestion and improves permitting prospects.

The business logic is therefore twofold: flexibility can generate direct market revenue, but it can also serve as a strategic asset for grid access and risk mitigation. This dual role strengthens the case for system-integrated operation in congested regions, yet it remains invisible to traditional efficiency metrics.

From a revenue perspective, flexibility can be decomposed into distinct products with different risk profiles: short-duration frequency regulation, reserve capacity, and longer-duration demand response or arbitrage. Each product entails specific telemetry and performance obligations, which affects participation costs and contractual design [13, 32, 25]. The operational implication is that a data center may need to dedicate specific assets to market

participation, which introduces opportunity costs that must be reflected in investment analysis.

Renewable integration occurs through both procurement and physical coupling. Many operators rely on corporate power purchase agreements or other instruments to match annual electricity use with renewable generation, while siting decisions increasingly favor regions with low-carbon supply and available network capacity. At the meso level, emerging "energy hub" concepts co-locate data centers with renewables, storage, and other flexible loads, using the data center as an anchor demand and a controllable participant in local balancing [45]. Demonstration projects in Nordic contexts also illustrate how clean generation and cold climates can support both lower overhead and novel energy-cascading uses of waste heat [44].

Smart local energy system concepts emphasize the value of coordinated, place-based integration that aligns generation, storage, and demand [37]. Data centers are well positioned to act as anchor loads within such systems because they provide stable demand and can deploy digital control for flexibility. Local energy communities and regional innovation ecosystems are increasingly exploring these configurations, but their success depends on governance arrangements that allocate benefits across utilities, municipalities, and operators. As a result, renewable coordination is as much an institutional and contractual challenge as a technical one.

From a business perspective, renewable coordination can reduce exposure to wholesale price volatility and strengthen ESG positioning, but it also binds projects to specific geographies and infrastructure timelines. In practice, the ability to co-locate with renewables is constrained by land availability and by the need for grid connection rights. This reinforces the meso-level spatial lock-in described above and underscores why renewable integration must be evaluated alongside land and grid constraints rather than as a purely procurement decision.

2.5. Cross-Level Misalignment and the KPI Gap

Integration is often slowed by cross-level misalignment. Macro policy encourages decarbonization and reporting; meso constraints demand grid relief and spatial coordination; micro finance depends on predictable cash flows. Conventional performance metrics, however, are largely internal-efficiency indicators that do not capture the system value of integration. PUE remains the dominant metric in reporting and industry practice, and standards such as ISO/IEC 30134 institutionalize it as a key benchmark [46, 17, 16].

Yet PUE classifies all non-IT electricity consumption as overhead, which means that integration assets—heat pumps, storage, controllable cooling—can worsen the metric even when they create external value. Empirical critiques show that PUE can misrepresent efficiency improvements and system contributions, particularly in settings where heat reuse or flexibility is material [20, 19, 47, 48].

This misalignment is amplified by policy-driven reporting. Directive (EU) 2023/1791 requires standardized energy performance reporting, including PUE and related indicators, and is complemented by the Commission's reporting requirements for data centers [10, 33]. These metrics have become governance instruments: they inform regulatory oversight, ESG reporting, and investment screening. When integration improves system outcomes but worsens PUE, the reporting framework can penalize projects that are otherwise beneficial, creating a mismatch between public objectives and private incentives.

For investors, this KPI gap is not merely semantic. Financing terms often embed performance expectations derived from standardized metrics, which means that projects can face higher discount rates or tighter covenants if they appear less efficient, even when they deliver system benefits. The result is a structural mispricing of system integration: the private cost of capital does not fully reflect public value creation. This tension is particularly acute in service-active regimes, where integration assets materially increase electricity consumption while reducing overall system costs. A metric that internalizes these external services is therefore a prerequisite for aligning governance signals with investment incentives [10, 11, 17].

An effective KPI for this context must satisfy two constraints simultaneously: backward compatibility with existing reporting frameworks and forward compatibility with system-value creation. Backward compatibility is required because regulatory reporting and industry benchmarks are built around PUE; forward compatibility is required because integration assets only become bankable when their system value is recognized in performance assessment. The siPUE metric introduced in Part II is designed as such a translation layer: it preserves the familiar PUE structure while crediting monetized system services. This allows the same indicator to speak to regulators, utilities, and financiers without erasing the business value of integration.

Figure 2 conceptualizes the macro–meso–micro decision domains that shape investment readiness. The circles represent domains of policy design, infrastructure coordination, and project execution. Overlapping areas denote

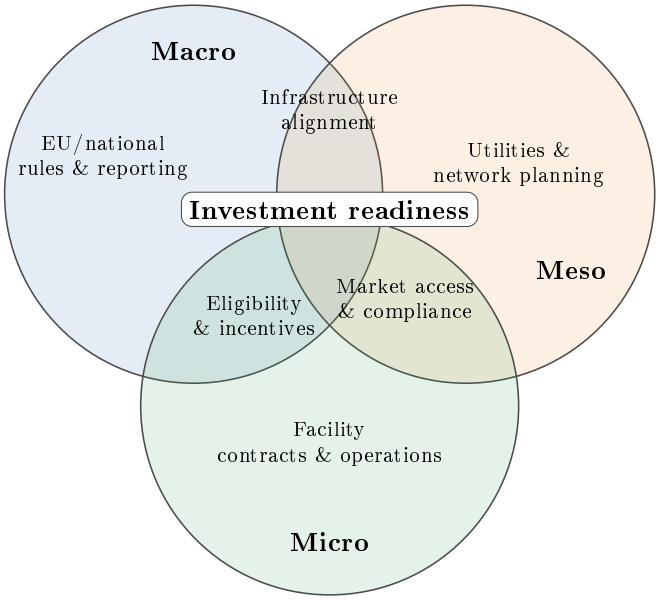


Figure 2: Conceptual macro–meso–micro governance structure. Investment readiness emerges when regulatory signals, infrastructure feasibility, and project-level execution align. This structuring is adapted from the multi-level perspective in socio-technical transitions literature [38] to fit the data-center energy-integration context

joint decision requirements rather than strict actor sets.

The macro–meso–micro framing used in Figure 2 aligns with broader smart local energy systems perspectives that emphasize actor coordination and local context [37]. Table 2 summarizes typical governance functions that frequently appear in investment due diligence for integrated data-center projects. These functions are the operational bridge between policy intent and project execution: they determine whether integration assets can be financed, permitted, and operated with reliable revenue streams.

The business implications of this cross-level misalignment are direct. A project that invests in integration assets can improve system outcomes and create long-term revenues, but still appear less efficient under conventional PUE reporting. This reduces coherence between profitability-driven investment decisions and KPI-centered reporting, particularly in service-active regimes where additional electricity use enables valuable system services. The analysis therefore motivates the need for a system-integrated performance indicator that preserves PUE comparability while internalizing external value.

The constraint set carried into Part II reflects this business context. De-

Table 2: Governance functions shaping investment readiness for system-integrated data centers.

Level	Typical functions relevant to investment readiness
Macro (EU/national)	Climate targets, reporting requirements, carbon-price exposure, and market-design rules for flexibility that shape expected cash flows and eligibility for services [8, 9, 10, 11, 42].
Meso (regional/local)	Network planning and connection rights, congestion management, district-heating access and tariff design, land availability, zoning and permitting, and infrastructure sequencing that determine feasibility and timing [7, 14, 15].
Micro (project)	Heat-offtake and flexibility contracts, metering and telemetry, operational control, and risk allocation between operator, utility, and third parties [15, 43, 13].

mand is bounded by the IT workload profile, revenue timing risk is represented through lifecycle construction-delay accounting, and electricity-side constraints are captured through the hourly energy balance and the three-node DC-OPF network model. Heat-side constraints combine heat-exchanger capacity, heat-pump capacity, recoverable heat, pipe limits, and district-heating acceptance. Market-access and technical limits on flexibility are represented by installed controllable capacity, while storage is constrained by power, energy, and state-of-charge dynamics. Land constraints are represented by capacity-linked site requirements and annualized lease or opportunity costs. Together, these constraints translate macro policy, meso infrastructure, and micro finance into the optimization problem used to define and evaluate siPUE.

The following section introduces siPUE within this techno-economic framework.

3. Part II: Methodological Contribution

Part II turns the business problem from Part I into a reproducible modeling exercise. The aim is not to forecast a particular site or to propose min–max regret as a new optimization method. The aim is narrower and more useful for the present argument: to define and test a PUE-compatible companion metric that distinguishes between ordinary facility overhead and electricity use that enables useful local-system services.

The framework has three design cases. The *no-services* case disables heat

export, flexibility, and storage services. It is the counterfactual in which siPUE must reduce to PUE. The *baseline* case represents a fixed service-active design, with incumbent levels of heat recovery, flexibility, battery, and efficiency investment. The *robust* case is an uncertainty-aware stress test for the metric. It is selected once for a set of uncertainty states using a normalized min–max regret criterion, then evaluated scenario by scenario to test whether siPUE reveals system value that PUE hides.

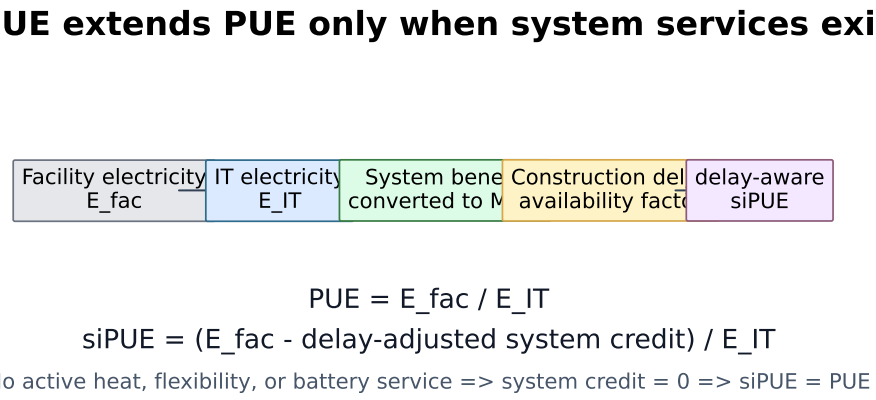


Figure 3: Metric logic for siPUE. PUE remains the internal energy ratio. siPUE subtracts a delay-adjusted, energy-equivalent credit for independently valued system benefits. When no services are active, the credit is zero and siPUE equals PUE.

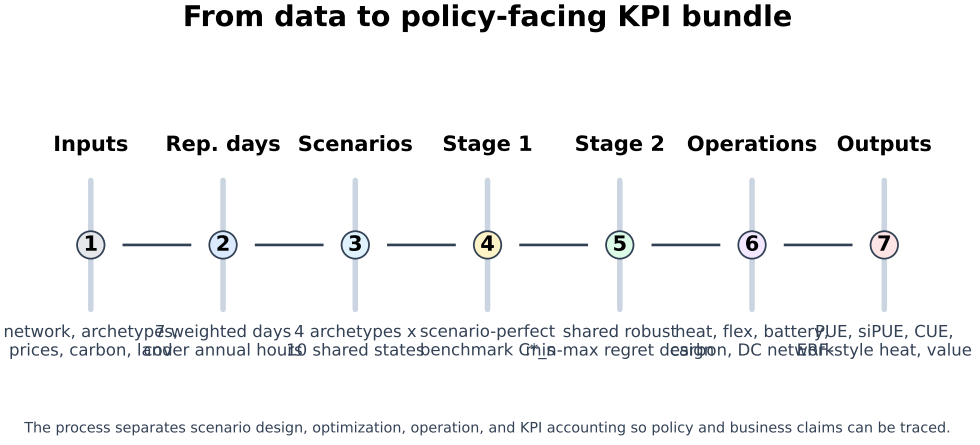


Figure 4: Model pipeline from inputs and scenarios to robust design, KPI outputs, validation diagnostics, and provenance tracking.

3.1. Definition and interpretation of siPUE

PUE is defined as

$$\mathrm{PUE} = \frac{E_{\mathrm{FAC}}}{E_{\mathrm{IT}}}, \tag{1}$$

where E_{FAC} is modeled facility electricity and E_{IT} is delivered IT electricity. The proposed siPUE keeps this denominator and adjusts only the numerator. It is a PUE-compatible companion metric that converts independently valued local-system benefits, including heat, flexibility, peak, carbon damage, and delay, into an energy-equivalent numerator credit. Let superscript 0 denote the no-services counterfactual and d denote a service-active design. The independent system-cost ledger is

$$C^d_{\mathrm{ind}} = C^d_{\mathrm{el}} + C^d_{\mathrm{trans}} + C^d_{\mathrm{peak}} + C^d_{\mathrm{CO2,dam}} - V^d_{\mathrm{heat}} - V^d_{\mathrm{flex}}, \tag{2}$$

where C_{el} is electricity resource cost, C_{trans} is transmission or network cost, C_{peak} is the independent peak-capacity cost, $C_{\mathrm{CO2,dam}}$ is the carbon-damage ledger, and V_{heat} and V_{flex} are independently valued avoided heat and flexibility costs. The net system benefit credited by siPUE is the counterfactual reduction in this independent ledger:

$$B^d_{\mathrm{sys}} = C^0_{\mathrm{ind}} - C^d_{\mathrm{ind}}. \tag{3}$$

Equivalently, positive terms in B_{sys} are avoided independent system costs, while negative terms are additional system burdens from the service-active design. No component is truncated before aggregation. A design with net negative B_{sys} therefore receives a negative energy credit and its siPUE increases relative to PUE; the outer $[\cdot]^+$ operator in Eq. 5 only prevents the adjusted numerator from becoming negative when net benefits are very large.

Table 3: Accounting boundary for the siPUE system-benefit term.

Included in B_{sys}	Excluded from B_{sys}
Electricity resource-cost changes, transmission or network-cost changes, independent peak-capacity effects, carbon-damage changes, avoided heat-system cost, and avoided flexibility-system cost.	Private heat and flexibility revenues, IT service revenue, ETS or carbon-market payments, capex, land lease or opportunity cost, private demand charges, taxes, subsidies, and financing terms.

This boundary keeps private settlement and public system value separate. Private heat and flexibility revenues appear in the market-cost objective, but they are not directly credited in siPUE. Conversely, carbon-market payments are kept in the private market ledger, while carbon damage enters only the independent system ledger. This separation avoids counting a private payment as a public system benefit or counting the same carbon effect twice.

The delay-adjusted energy credit is

$$E_{\text{credit}} = a_s \frac{B_{\text{sys}}}{p_{\text{ref}}}, \tag{4}$$

where p_{ref} is the calibrated electricity reference price in EUR/MWh and $a_s \in [0, 1]$ is the lifecycle availability factor after construction delay. The reference price is a unit converter, not a fitted performance parameter. In the canonical run it is fixed before optimization at 56.63 EUR/MWh, based on the no-services electricity-resource marginal cost at the data-center node. ETS payments and carbon damage are excluded from p_{ref} because those terms are already handled in separate ledgers. Freezing p_{ref} across all designs and uncertainty states prevents the metric from improving simply because the conversion factor changes.

siPUE is

$$\text{siPUE} = \frac{[E_{\text{FAC}} - E_{\text{credit}}]^+}{E_{\text{IT}}}, \tag{5}$$

with $[z]^+ = \max(0, z)$. Four internal property checks are required before interpretation: the metric is nonnegative, it reduces to PUE when no services are active, positive system benefit lowers siPUE, and construction delay weakens the credit. The regenerated property table satisfies these checks.

3.2. Reference-price sensitivity and anti-gaming rule

The reference price is the main conversion choice in siPUE, so it is treated as a first-order sensitivity rather than a hidden calibration. Holding design and operation fixed, a multiplier k gives

$$\text{siPUE}(k) = \frac{[E_{\text{FAC}} - a_s B_{\text{sys}}/(k p_{\text{ref}})]^+}{E_{\text{IT}}}. \tag{6}$$

The anti-gaming rule is that p_{ref} must be selected before the service-active design is evaluated. In this study it is the frozen no-services electricity-resource marginal cost at the data-center node. It is not optimized, not

Table 4: Reference-price sensitivity for the canonical robust rows. Dispatch and ledgers are held fixed; only the value-to-energy conversion factor changes.

Reference-price case	Interpretation	Average siPUE
$0.5p_{\text{ref}}$	Low conversion price; stronger energy credit	1.019
$1.0p_{\text{ref}}$	Canonical frozen reference price	1.216
$1.5p_{\text{ref}}$	High conversion price; weaker energy credit	1.282

selected by the operator after seeing the result, and not allowed to vary across designs or uncertainty states. ETS payments and carbon damage are excluded from p_{ref} because those effects already enter separate ledgers. The sensitivity in Table 4 recomputes only Eq. 6; dispatch, heat delivery, emissions, PUE, and the sign of B_{sys} are unchanged.

This sensitivity is intentionally large enough to report: the reference price affects the magnitude of the siPUE credit, even though it does not change the underlying physical or monetary ledgers. For that reason, siPUE should always be reported with the reference-price rule and a sensitivity band.

siPUE is therefore not a thermodynamic efficiency ratio and should not replace PUE. It is a system-adjusted performance indicator. For operators and investors, it gives a KPI view of external value that can be read next to PUE. For policy users, it gives a way to discuss heat recovery and flexibility without treating every MWh of additional facility electricity as waste.

3.3. Two-stage design model

Let $s \in \mathcal{S}$ index the shared uncertainty states, x denote design variables, and y_s denote hourly operational variables. Stage 1 solves a scenario-perfect benchmark:

$$C_s^\star = \min_{x,y_s} C_s(x, y_s), \tag{7}$$

where C_s is lifecycle market cost for scenario s. Stage 2 chooses one shared design $\hat{x}$ for the uncertainty set. Regret is computed against the scenario-perfect benchmark:

$$R_s(\hat{x}) = \frac{C_s(\hat{x}, y_s) - C_s^\star}{\max(\epsilon, C_s^\star)}. \tag{8}$$

The robust design is selected in two passes: the first minimizes the worst normalized regret across scenarios, giving $Z^\star$ from Eq. 8; the second fixes $Z \leq Z^\star(1+\tau)$ and, within that near-worst-case band, minimizes mean life-cycle cost across scenarios. In the canonical run $\tau = 10^{-7}$, so this tie-break only resolves numerically degenerate ties at the worst-case-regret optimum – it never trades a material amount of regret for lower average cost. The canonical result bundle reports only the no-services, baseline, and robust design cases; scenario-perfect solutions serve as planning benchmarks and validation references, not as additional canonical design cases.

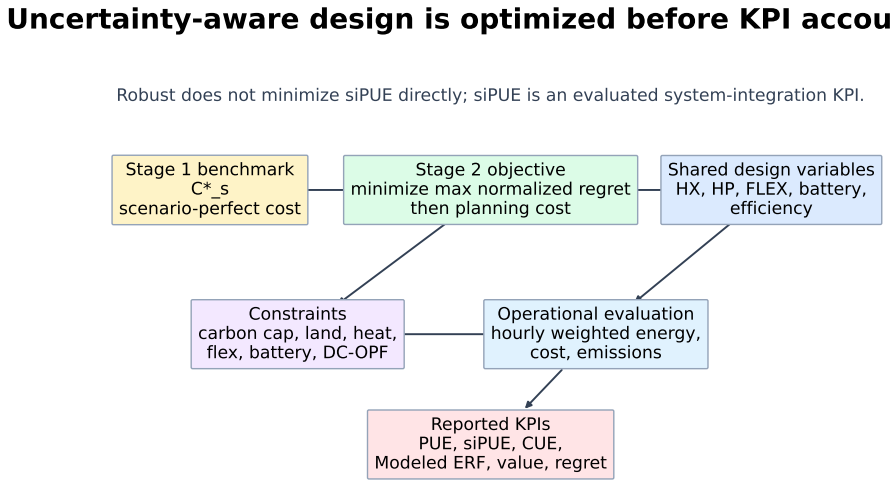


Figure 5: Optimization map linking design variables, scenario-perfect benchmarks, normalized regret, and physical constraints.

3.4. Operational constraints and ledgers

Hourly operation is represented as a linear program. The facility energy balance links grid import, battery charging and discharging, IT load, cooling overhead, and heat-pump electricity. Heat export is bounded by heat-exchanger capacity, heat-pump capacity, recoverable heat, pipe limits, and district-heating acceptance. Flexibility is bounded by installed controllable capacity and service availability. Battery operation follows state-of-charge dynamics with charge and discharge limits.

The electricity system is represented by a stylized three-node DC-OPF. Thermal generation, renewable availability, line capacities, and nodal settlement prices determine the cost of supplying the data center and the wider

network. This network model is useful for consistent accounting, but the current three-node representation should be read as a stylized system context rather than a site-specific grid study.

Carbon is kept in separate ledgers. Market carbon cost enters the private market ledger. Carbon damage enters the independent system-cost ledger. The two are reported together, but they are not merged into a single private-credit claim. Land is also stylized: it is modeled as lease or opportunity-cost accounting tied to core facility, cooling, heat, flexibility, and battery footprints. It is a useful constraint for comparing designs, but it is not a physical siting backstop.

3.5. Data and scenario design

The model uses a calibrated three-node hourly time series with 8,760 sequential hours. The annual run is compressed into seven representative days. Hourly weights sum to 8,760 and day weights sum to 365, so operating quantities are annualized once. Fixed annual terms, such as capex, demand charges, and land lease cost, are not multiplied again by representative-day weights.

The scenario set has four archetypes and ten shared uncertainty states, giving 40 canonical rows. The archetypes are AI/HPC, hyperscale cloud, colocation, and enterprise/edge. Each uncertainty state changes service prices, heat conditions, construction delay, carbon caps, and related operating conditions using seeded scenario construction. The design and KPI outputs are normalized by IT energy or IT capacity where appropriate, so capacity comparisons are interpretable across archetypes.

Table 5: Canonical scenario structure used in the regenerated evidence bundle.

Dimension	Count	Role in the model
Archetypes	4	AI/HPC, hyperscale cloud, colocation, enterprise/edge
Shared uncertainty states	10	Heat, flexibility, carbon-cap, delay, and service conditions
Canonical rows	40	One row for each archetype and uncertainty-state pair
Canonical design cases	3	no-services, baseline, robust
Representative days	7	Weighted annual operating approximation

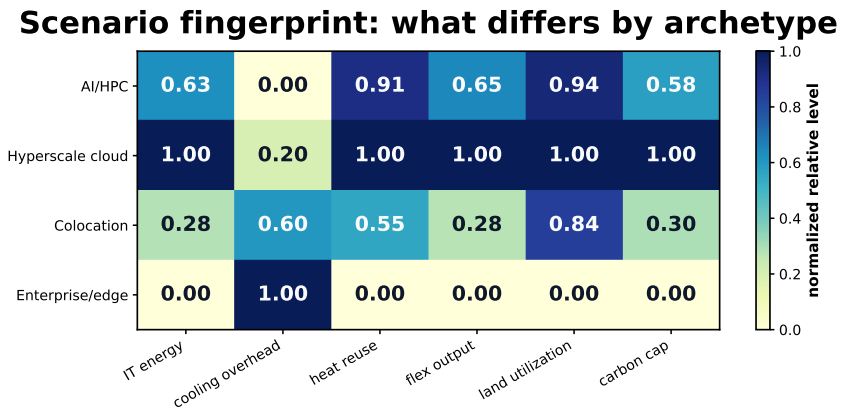
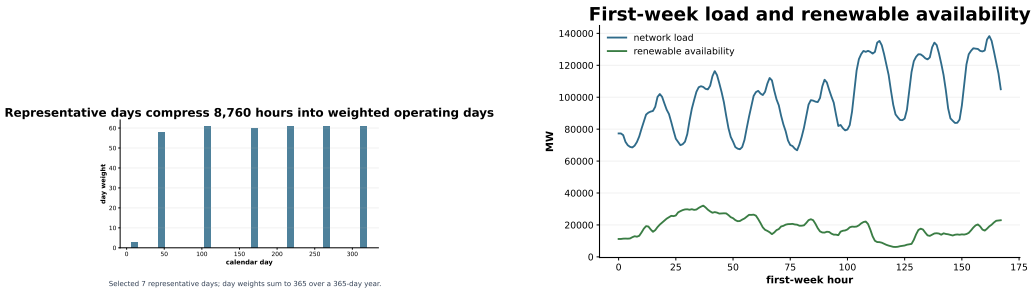


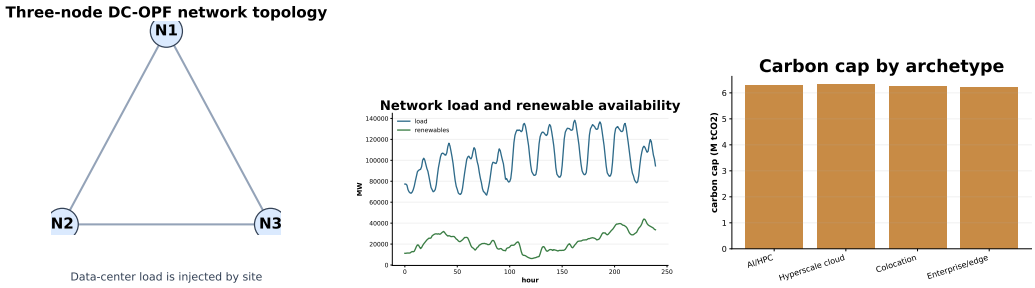
Figure 6: Input scenario fingerprint. Values are normalized within each column, so the figure shows relative scenario structure rather than absolute engineering limits.



(a) Selected representative days and weights.

(b) First-week load and renewable profiles.

Figure 7: Representative-day compression and hourly system context.



(a) Network topology.

(b) Load and renewables.

(c) Carbon-cap context.

Figure 8: Stylized electricity-network and carbon context used by the operational model.

4. Results and Discussion

4.1. PUE and siPUE move differently

Table 6 reports average modeled metrics from the canonical output. The no-services case gives the required identity: PUE and siPUE are both 1.218 on average. The baseline design raises average PUE to 1.277 while lowering average siPUE to 1.193. The robust design raises average PUE further to 1.413, while average siPUE remains close to the no-services value at 1.216. This is the main result. System-integrated designs can look worse under PUE because integration equipment uses electricity, while siPUE reveals system value that PUE hides.

Table 6: Compact metric benchmark, averaged across 40 canonical rows. Carbon, heat, and emissions are kept as separate ledgers rather than re-labelled as standard KPIs.

Design case	PUE	siPUE	Post-construction siPUE	System benefit (EUR/MWh IT)
No-services	1.218	1.218	1.218	0.000
Baseline	1.277	1.193	1.190	4.728
Robust	1.413	1.216	1.209	11.155

The direction is not uniform across all scenarios, and it should not be presented as a pass/fail score. A clearer presentation is a directional count relative to the no-services counterfactual. PUE is lower than no-services in none of the baseline or robust rows. siPUE is lower than no-services in 36 of 40 baseline rows and 23 of 40 robust rows. The robust design therefore does not make siPUE universally lower than the no-services case. It does show that PUE and siPUE answer different questions once integration assets are installed.

Table 7: Directional comparison against the no-services counterfactual. These counts are descriptive, not a pass/fail test.

Archetype	Baseline siPUE lower	Robust siPUE lower	Baseline PUE lower	Robust PUE lower
AI/HPC	10/10	6/10	0/10	0/10
Hyperscale cloud	7/10	8/10	0/10	0/10
Colocation	10/10	5/10	0/10	0/10
Enterprise/edge	9/10	4/10	0/10	0/10
All rows	36/40	23/40	0/40	0/40

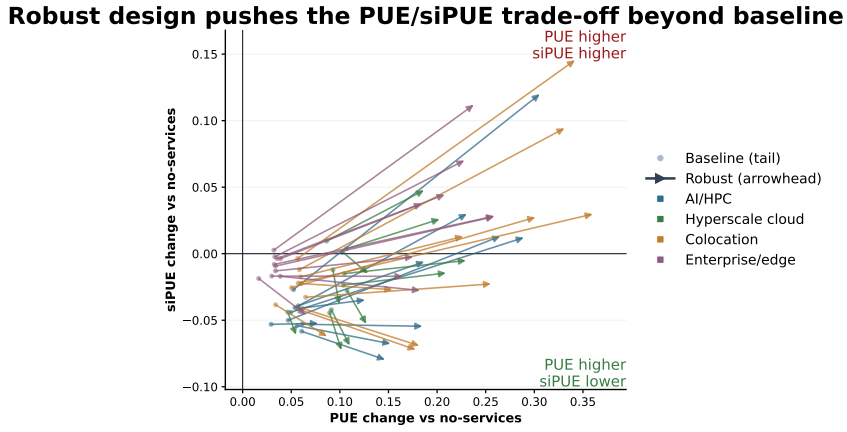


Figure 9: PUE and siPUE tradeoff compass. Each arrow moves one scenario from the baseline design to the robust design, both measured as change relative to no-services.

4.2. Credit decomposition

Table 8 decomposes the robust siPUE credit into the independent ledger components that create or reduce local-system value. Positive entries reduce siPUE because they lower independent system cost relative to the no-services counterfactual. Negative entries raise siPUE because they are additional system burdens from the service-active design. The robust design creates large heat, flexibility, and carbon-damage benefits, but those benefits are partly offset by additional electricity-resource, peak-capacity, and transmission burdens. Construction delay then weakens the credit before it enters the siPUE numerator.

This ablation is the accounting reason siPUE remains close to the no-services value while PUE worsens. The service-active design uses substantially more facility electricity, so PUE rises. siPUE does not erase that burden; it offsets part of it with independently valued local-system benefits that PUE cannot observe.

4.3. Design portfolios and system value

The robust design is a different technology portfolio, not a lower-PUE tuning of the same system. Figure 10 shows the capacity intensity of heat recovery, heat pumps, flexibility, and batteries by archetype. Figure 11 decomposes the resulting system value. These figures keep the economic story

Table 8: Component ablation of the robust siPUE credit, averaged across 40 canonical rows. Values are EUR/MWh IT. Components above the delay row are before lifecycle delay; the final row is the delay-adjusted benefit credited by siPUE.

Component	Contribution	Interpretation
Heat system value	12.268	Avoided independent heat-system cost
Flexibility system value	3.346	Avoided independent flexibility cost
Carbon damage	6.546	Avoided carbon-damage ledger cost
Peak capacity	-1.240	Additional independent peak-capacity burden
Electricity resource cost	-8.926	Additional power-system resource burden
Transmission / network cost	-0.418	Additional network-flow burden
Undelayed system benefit	11.575	Net benefit before delay adjustment
Construction-delay adjustment	-0.420	Lost credit from delayed availability
Delay-adjusted system benefit	11.155	Benefit entering the siPUE numerator

visible: higher facility energy can be a cost of delivering heat and flexibility services, but the benefit depends on the local price, heat, land, and delay conditions.

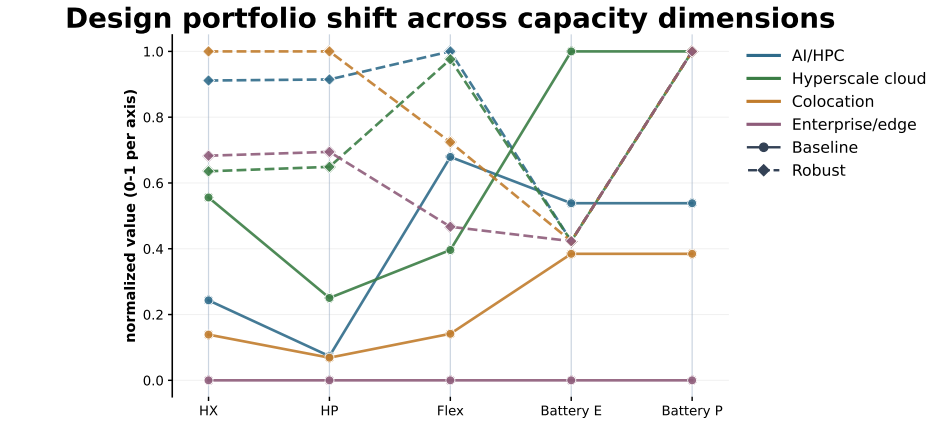


Figure 10: Baseline and robust design portfolios by archetype, normalized per MW IT.

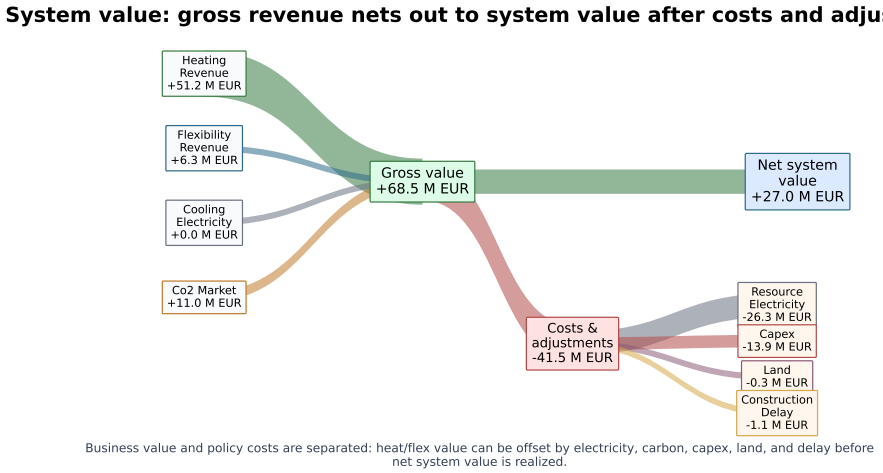


Figure 11: System-value flow for the robust design. Heat and flexibility benefits are reported separately from electricity, carbon, capex, land, and delay adjustments.

4.4. Carbon and delay accounting

The model reports emissions, market carbon cost, carbon damage, avoided heat emissions, and power emissions as separate ledgers. This separation is

important because carbon-damage valuation can change monetized system benefit without changing physical emissions. For that reason, carbon damage is retained as a ledger item, but it is not used as one of the main physical sensitivity factors.



(a) Power emissions.

(b) Avoided heat emissions.

Figure 12: Carbon-ledger separation in the robust design.

Construction delay affects siPUE through the availability factor in Eq. 4. It reduces the credit for useful system benefit while leaving the underlying facility energy accounting intact. The canonical outputs distinguish post-construction siPUE from delay-adjusted siPUE for this reason.

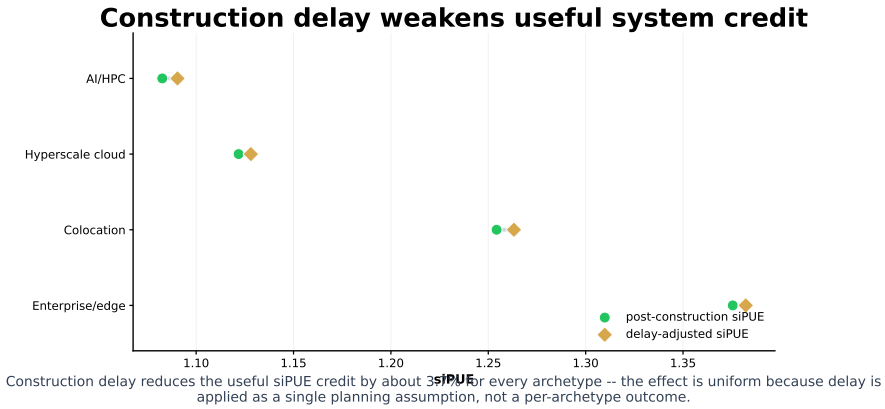


Figure 13: Bridge from post-construction siPUE to delay-adjusted siPUE. Delay weakens the useful system credit before it enters the metric.

4.5. Validation and reproducibility diagnostics

The validation results should be read as internal reproducibility diagnostics. They do not externally validate siPUE against observed projects. The property checks verify metric behavior: nonnegativity, no-services identity,

delay weakening of credit, positive-benefit monotonicity, and ledger separation. The train/test diagnostic trains on earlier uncertainty states and evaluates held-out states. The out-of-sample diagnostic evaluates newly sampled states against the robust design.

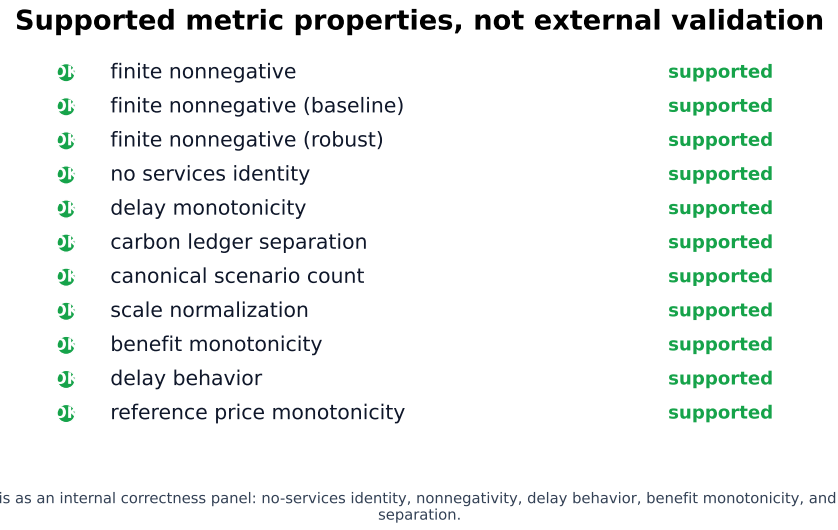


Figure 14: Internal metric property checks. These are correctness checks, not external validation claims.

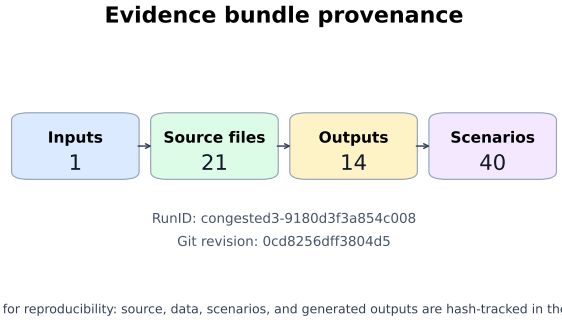


Figure 15: Provenance map summarizing hashed inputs, source files, and outputs tracked for the canonical run, supporting the reproducibility framing of the diagnostics above.

The canonical metric benchmark, the train/test diagnostic, and the out-of-sample diagnostic are regenerated together from the same source tree for this paper. Each nonetheless carries its own run identifier, because that

identifier hashes the scenario set together with the source code, and the three diagnostics deliberately cover different scenario sets: the full 40-row canonical set, the 7-state training fold, and independently sampled out-of-sample draws. The differing identifiers reflect that intended difference in scenario coverage, not a difference in source code version.

Table 9: Validation diagnostics from the regenerated output bundle. Regret is reported in EUR/MWh IT where comparable.

Diagnostic	Rows	Feasible rows	Mean regret	Regret range
Train/test held-out	12	12	1.036	0.003 to 3.759
Out-of-sample	40	40	0.429	0.012 to 1.241

The higher mean and wider range for train/test regret reflects how that diagnostic is constructed: the underlying design is trained on only 7 of the 10 canonical uncertainty states and evaluated against the 3 withheld states, including the highest-value "service-upside" state, which the design most underprovisions for. The out-of-sample diagnostic instead evaluates the design trained on the full 10-state set against independently sampled draws that rarely combine the same extremes, giving a lower and more stable regret.

Is cost-regret minimization leaving siPUE value on the table?. Stage 2 selects the robust design by minimizing worst-case normalized *cost* regret (Eq. 8); siPUE is never a Stage 2 objective, only a quantity computed on the resulting design. This raises a direct question: would a design chosen to minimize siPUE regret instead of cost regret look different, and does the cost-optimal robust design leave siPUE headroom unexploited? Table 10 answers this using the same scenario-perfect solutions already computed as the Stage 1 benchmark: for each of the 40 canonical rows, siPUE regret is defined as siPUE of the shared robust design minus siPUE of the scenario-specific, cost-minimizing perfect-foresight design for that row.

Table 10: siPUE regret of the robust design relative to the scenario-perfect (cost-minimizing, perfect-foresight) benchmark, by archetype. Regret is $\text{siPUE}_{\text{robust}} - \text{siPUE}_{\text{scenario-perfect}}$ (dimensionless); negative values mean the shared robust design already achieves a lower, more favorable siPUE than the scenario-specific cost optimum for that row.

Archetype	Rows	Mean regret	Regret range
AI/HPC	10	−0.0072	−0.0249 to 0.0037
Hyperscale cloud	10	−0.0038	−0.0139 to 0.0002
Colocation	10	0.0005	−0.0105 to 0.0092
Enterprise/edge	10	0.0026	−0.0006 to 0.0129
All rows	40	−0.0020	−0.0249 to 0.0129

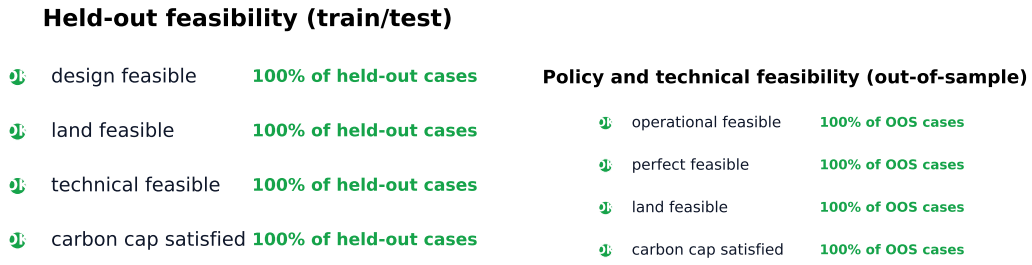
Across all 40 rows, siPUE regret is small in absolute terms (at most 0.013 siPUE units, under 0.9% of the canonical robust siPUE of 1.216) and centered near zero: two archetypes show a small negative mean regret, meaning the cost-optimal robust design's siPUE is, on average, already at or slightly below what a scenario-specific cost optimum would achieve, while the other two show a small positive mean regret. Because this diagnostic never rewards siPUE directly, a mean regret this close to zero is evidence that minimizing cost regret does not leave material siPUE value unexploited in this evidence bundle, not a claim that the two objectives are formally equivalent. This diagnostic bounds the size of any gap a siPUE-regret Stage 2 objective could close, but it does not itself construct that alternative design.

Constructing the alternative directly. We therefore built the alternative rather than only bounding it. Stage 2 was re-solved with worst-case *siPUE* regret against a per-scenario siPUE-perfect benchmark as the primary criterion, replacing cost regret in Eq. 8, with the same lexicographic cost tie-break so the two runs are comparable side by side and differ only in the primary worst-case criterion. This required a matching Stage 1 siPUE-perfect benchmark, solved by minimizing siPUE (instead of lifecycle cost) for each scenario in isolation, using the same independent-system-cost ingredients as the siPUE credit itself so the two stages cannot drift apart. Table 11 compares the resulting shared design against the canonical cost-regret robust design for the same 40 rows.

Table 11: Cost-regret-minimizing (canonical robust) design versus siPUE-regret-minimizing design, by archetype. Worst-case regret is the maximum, within each archetype's shared 10-scenario group, of siPUE for that design minus the scenario-perfect siPUE benchmark. Average cost is the mean lifecycle market cost (NPV, EUR) across the same 10 rows.

Archetype	Worst regret (cost-regret)	Worst regret (siPUE-regret)	Avg. cost difference	Design identical?
AI/HPC	0.0475	0.0496	−0.001%	No
Hyperscale cloud	0.0233	0.0233	0.00%	Yes
Colocation	0.0403	0.0403	0.00%	Yes
Enterprise/edge	0.0195	0.0195	0.00%	Yes

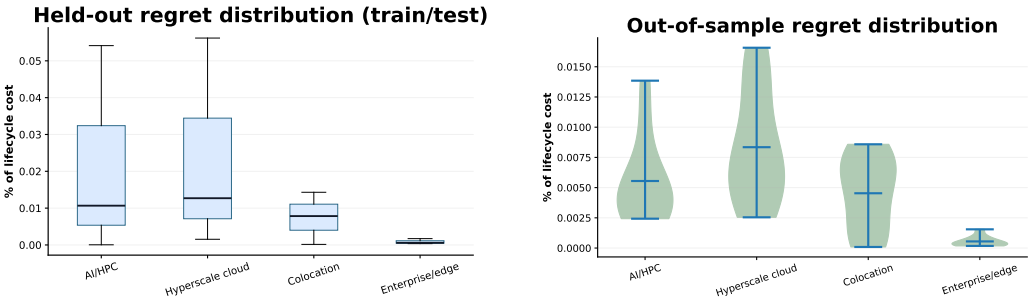
For three of the four archetypes, minimizing worst-case siPUE regret directly selects the *exact same* shared design, capacity for capacity, as minimizing worst-case cost regret: identical worst-case regret and identical average lifecycle cost across all 10 rows in each group. For AI/HPC, the two objectives diverge, but by an amount consistent with solver numerical tolerance at this problem's scale rather than a substantive design difference: the siPUE-regret design's own worst-case regret is 0.002 siPUE units *higher* than the cost-regret design achieves on the same benchmark, while its average lifecycle cost is 0.001% *lower*; a genuine trade-off would show the siPUE-regret design winning on siPUE and losing on cost, not losing on both its own criterion and matching in magnitude something that small against a lifecycle cost measured in tens of billions of EUR. Averaged across all 40 rows, mean siPUE is 1.2160 under cost-regret minimization and 1.2164 under siPUE-regret minimization, and mean PUE is 1.4130 versus 1.4140. Both differences are an order of magnitude smaller than the headline no-services-to-robust shift reported in Table 6. Combined with the bound in Table 10, this directly constructed alternative confirms rather than merely bounds the earlier finding: the paper's canonical robust design, obtained by minimizing worst-case cost regret alone, is already the same design (or economically indistinguishable from it) that a siPUE-regret objective would select. Reformulating Stage 2 around siPUE would not have changed the reported results.



(a) Train/test feasibility.

(b) Out-of-sample feasibility.

Figure 16: Feasibility diagnostics for held-out and out-of-sample cases.



(a) Train/test regret.

(b) Out-of-sample regret.

Figure 17: Regret diagnostics for held-out and out-of-sample cases, complementing the feasibility panels above and the summary statistics in Table 9.

4.6. Sensitivity analysis

The manuscript sensitivity varies five factors: heat price, flexibility price, cooling overhead, land availability, and construction delay. Carbon damage is excluded from this sensitivity set because it is a valuation ledger, not a physical design lever. Congestion, battery-ceiling, and regret-surface diagnostics are also excluded from the final manuscript-support bundle.

Table 12: All-archetype sensitivity results for the robust design. Values are averages across the 40-row scenario set for each sensitivity case.

Factor	Level	PUE	siPUE	Benefit (EUR/MWh IT)	Land use (%)
Canonical	base	1.413	1.216	11.155	67.3
Heat price	low	1.311	1.178	7.554	54.4
Heat price	high	1.434	1.226	11.812	70.0
Flexibility price	low	1.413	1.228	10.469	65.6
Flexibility price	high	1.412	1.214	11.223	67.9
Cooling overhead	low	1.365	1.171	10.981	66.2
Cooling overhead	high	1.461	1.261	11.326	68.4
Land availability	low	1.390	1.222	9.530	87.6
Land availability	high	1.413	1.216	11.155	56.1
Land availability	unconstrained	1.413	1.216	11.155	0.7
Construction delay	0 months	1.414	1.209	11.639	67.9
Construction delay	12 months	1.413	1.225	10.603	67.0

The largest siPUE response is cooling overhead, followed by heat price. Construction delay mainly works through the lifecycle credit. Flexibility price changes the modeled benefit more than it changes PUE, which is consistent with a value channel rather than a large physical-energy channel. Low land availability binds in three all-scenario robust cases and reduces system benefit, but the land model remains stylized.

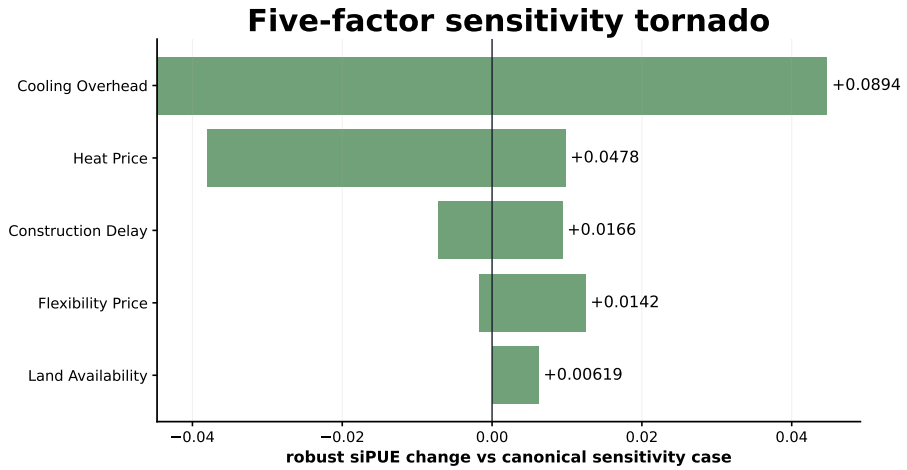


Figure 18: Five-factor sensitivity tornado for robust siPUE. Carbon damage is not included as a sensitivity factor.

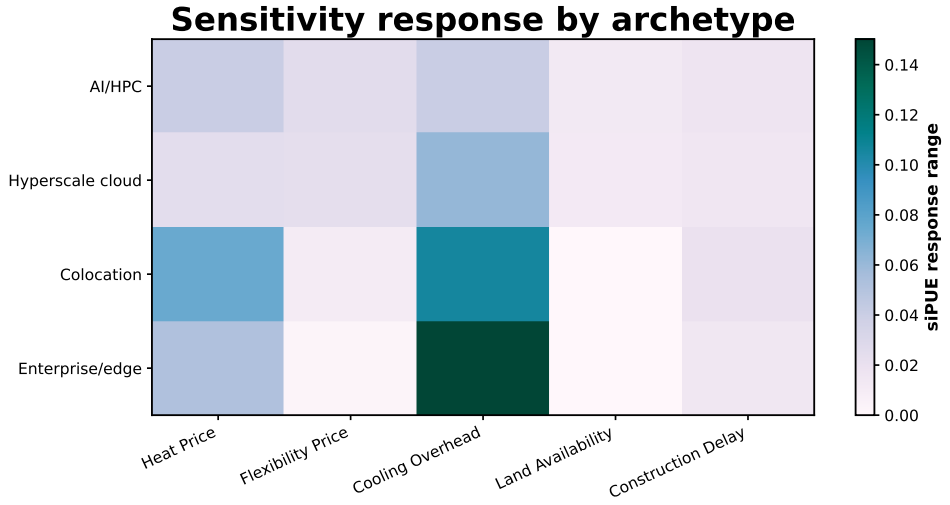


Figure 19: Sensitivity response by archetype. The figure shows response ranges, not site-specific forecasts.

4.7. Face validity against published benchmarks

The diagnostics above are internal. This subsection adds a narrower, external check: whether the model's modeled PUE and heat-reuse levels are

in a plausible range next to published, real-project figures. This is a face-validity comparison, not a calibration or a statistical validation exercise, and no model parameter is fit to the figures below.

For PUE, the Uptime Institute's 2024 global survey reports an industry-average PUE of 1.56–1.58, with hyperscale operators (Google, Amazon Web Services, Microsoft) claiming 1.2 or lower at some sites, and new-build large colocation campuses commonly targeting around 1.4 [49]. The no-services (pure facility-overhead) PUE in the canonical archetypes is 1.10 for AI/HPC and 1.15 for Hyperscale cloud, both inside the published hyperscale range. Colocation (1.26) and Enterprise/edge (1.36) sit below the published industry-wide average of 1.56–1.58, closer to what only modern, newly built facilities achieve. This is a deliberate scope choice, not an error: the archetypes represent facilities capable of considering integration investment, which skews toward the newer, more efficient end of the real fleet, and the comparison should not be read as claiming the archetypes represent the average colocation or enterprise site in operation today.

For heat reuse, the Mäntsälä data center in Finland operates a dedicated 4.0 MW heat pump and reuses 75% of its energy as district heat, a real, purpose-built, single-service case [50]. In this study, delivered useful heat divided by modeled facility electricity reaches 0.30–0.33 for AI/HPC and Colocation and 0.22–0.26 for Hyperscale cloud and Enterprise/edge, well below the Mäntsälä figure. This ratio is used only as a face-validity output, not as an official Energy Reuse Factor claim. The modeled archetypes are general-purpose facilities that size heat recovery as one constrained service among several – heat, flexibility, storage, land, and construction-delay exposure all compete for the same capital – rather than single-purpose heat-export sites, so the gap reflects that narrower mandate rather than a modeling shortfall. The recoverable-heat-fraction inputs (0.20–0.50 depending on archetype) cap what any design in this study could reach even before capacity and cost trade-offs bind.

Table 13 summarizes both comparisons. Both directions of mismatch (PUE running better than the published average; heat-delivery intensity running below a dedicated best-practice site) are consistent with the archetypes' intended scope, but they bound how far the numerical results in this paper should be extrapolated to a specific real facility.

Table 13: Face validity of modeled PUE and heat-reuse levels against published, real-project figures. This is a plausibility check, not a calibration.

Quantity	Published reference	Modeled range (this study)
Hyperscale/AI PUE	1.1–1.2 at leading sites [49]	1.10–1.15 (no-services)
Colocation/enterprise PUE	1.4 (new-build) to 1.56–1.58 (industry average) [49]	1.26–1.36 (no-services)
Delivered useful heat / facility electricity	0.75 at a dedicated site (Mäntsälä) [50]	0.22–0.33 (robust design output)

4.8. Limitations and future work

Several limitations define the scope of the results. The scenarios are stylized and should not be treated as site forecasts. The three-node network is useful for consistent DC-OPF accounting, but it is not a replacement for a local grid interconnection study. The land constraint is a lease or opportunity-cost representation, not a reliable physical siting backstop. Some archetype-specific conclusions remain evidence gaps, especially hyperscale flexibility, enterprise/edge flexibility, and enterprise/edge heat recovery.

The current model does not include water use, embodied carbon, formal Renewable Energy Factor, or official ISO Energy Reuse Factor accounting. Section 4.7 adds a face-validity comparison against published PUE and heat-reuse figures, but this is a plausibility check against aggregate published figures, not calibration or transaction-level validation. Future work should add observed project data at the transaction level, richer network models, water and embodied-carbon ledgers, contracting structures between data centers and utilities, and external validation against measured heat and flexibility transactions.

5. Conclusion

This paper connects a policy and market synthesis of data centers in local energy systems with a reproducible method for evaluating system-integrated designs. The proposed siPUE indicator is a PUE-compatible companion

metric that converts independently valued local-system benefits, including heat, flexibility, peak, carbon damage, and delay, into an energy-equivalent numerator credit. The metric reduces to PUE when no services are active, which preserves backward compatibility with existing reporting.

The regenerated 40-row canonical evidence bundle supports a conservative claim. PUE and siPUE can move in opposite directions under system-integrated designs. Average PUE rises from 1.218 in the no-services case to 1.413 in the robust case. Average siPUE remains close to the no-services value, moving from 1.218 to 1.216, because the robust design creates modeled system benefit that offsets part of its higher facility electricity use. The baseline design has a lower average siPUE of 1.193, but it also provides less modeled system benefit and lower heat-reuse intensity than the robust design.

The compact metric benchmark shows why siPUE should be reported beside PUE rather than replacing it. PUE measures internal facility overhead. Carbon, heat, and emissions remain separate ledgers. siPUE translates independently valued system benefit into the same energy-normalized language as PUE.

Validation diagnostics support internal consistency rather than external validation. The no-services identity, nonnegativity, delay behavior, benefit monotonicity, and ledger separation checks are satisfied. Train/test and out-of-sample diagnostics remain feasible and show nonnegative comparable regret. A further diagnostic compares the robust design's siPUE against the scenario-perfect, cost-minimizing benchmark for each row; the resulting siPUE regret is small (at most 0.013 siPUE units, under 0.9%) and centered near zero, indicating that minimizing worst-case *cost* regret in Stage 2 does not leave material siPUE value unexploited relative to a design chosen with foresight of the scenario, even though siPUE is never a Stage 2 objective. We tested this directly by re-solving Stage 2 with worst-case siPUE regret, rather than cost regret, as the primary criterion: for three of four archetypes the two objectives selected the identical shared design, and the fourth diverged by an amount consistent with solver numerical tolerance at this problem's scale rather than a substantive trade-off. Reformulating Stage 2 around siPUE would not have changed the reported results. A face-validity comparison against published PUE and heat-reuse figures places the modeled ranges inside or near the published range for hyperscale facilities, on the more efficient side of the published range for colocation and enterprise sites, and below a dedicated best-practice heat-reuse site, consistent with the archetypes' intended scope; this is a plausibility check, not a calibration. The sensitivity

analysis suggests that cooling overhead, heat price, construction delay, flexibility price, and land availability all affect the robust design in interpretable ways, while carbon-damage valuation is better treated as a separate policy ledger.

The practical implication is simple: a data center can look worse under PUE when it becomes more useful to the local energy system. That does not make PUE wrong. It means PUE is incomplete for service-active designs. siPUE reveals system value that PUE hides in those cases, with explicit limits, auditable inputs, and a clear no-services counterfactual.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data and results underlying this study are available in the same BRAVE GitHub repository as the code (see Code availability below).

Code availability

The computational implementation supporting this study is available in the BRAVE GitHub repository: https://github.com/sambeets/BRAVE/tree/main/sipue_modelling.

Acknowledgement

This research has received financial support from Interreg project BRAVE, and FME RenewHydro (Grant 350468, Research Council of Norway).

References

[1] E. Commission, Energy-efficient cloud computing technologies and policies for an eco-friendly cloud market – Final study report, Technical Report, Publications Office of the European Union, 2020. URL: https://data.europa.eu/doi/10.2759/3320. doi:10.2759/3320.

[2] ICIS, 2024, Europe data centre power demand, URL: https://www.icis.com/explore/resources/data-centres-hungry-for-power/.

[3] International Energy Agency, 2023, Data centres & networks, URL: https://www.iea.org/energy-system/buildings/data-centres-and-data-transmission-networks.

[4] World Economic Forum, 2025, Artificial intelligence's energy paradox: Balancing challenges and opportunities, URL: https://www.weforum.org/.

[5] oEnergetice.cz, 2024, Report oenergetice.cz 2024, URL: https://oenergetice.cz/energostat/report/rocni/2024.

[6] Elia Group, 2025, Integrated annual report 2024: Securing clean competitiveness, URL: https://investor.eliagroup.eu/en/reports-and-results/reports-for-elia-group.

[7] T. Joon, 2024, Data centres in ireland, URL: https://publicpolicy.ie/papers/data-centres-in-ireland/.

[8] European Commission, 2021, Communication from the commission to the european parliament, the council, the european economic and social committee and the committee of the regions: "fit for 55": delivering the eu's 2030 climate target on the way to climate neutrality, URL: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52021DC0550.

[9] European Commission, 2022, Communication from the commission to the european parliament, the council, the european economic and social committee and the committee of the regions: Digitalising the energy system – eu action plan, URL: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52022DC0552.

[10] European Union, 2023, Directive (eu) 2023/1791 of the european parliament and of the council of 13 september 2023 on energy efficiency and amending regulation (eu) 2023/955 (recast), URL: http://data.europa.eu/eli/dir/2023/1791/oj.

[11] Climate Neutral Data Centre Pact, 2025, Climate neutral data centre pact – the green deal needs green infrastructure, URL: https://www.climateneutraldatacentre.net/.

[12] M. Wahlroos, M. Pärssinen, S. Rinne, S. Syri, J. Manner, Future views on waste heat utilization – case of data centers in northern europe, Renewable and Sustainable Energy Reviews 82 (2018) 1749–1764. doi:10.1016/j.rser.2017.10.058.

[13] I. Alaperä, S. Honkapuro, J. Paananen, Data centers as a source of dynamic flexibility in smart grids, Applied Energy 229 (2018) 69–79. doi:10.1016/j.apenergy.2018.07.056.

[14] Euroheat & Power, Harnessing waste heat from data centres for sustainable urban development, 2023. URL not available.

[15] Bytes2Heat, Best-Practice-Overview: Learning from Inspiring Projects for Waste Heat Recovery from Data Centers, Technical Report, Bytes2Heat, 2024. URL: https://www.bytes2heat.com/static/files/20240403_BP%C3%9C_V2404_eng.pdf, accessed: 2025-11-25.

[16] C. Belady, M. Patterson, V. Avelar, Green Grid Data Center Power Efficiency Metrics: PUE and DCiE, Technical Report, The Green Grid, 2008. URL: https://www.thegreengrid.org.

[17] The Green Grid, PUE: A comprehensive examination of the metric, 2014. URL: https://datacenters.lbl.gov/resources/pue-comprehensive-examination-metric, available at https://datacenters.lbl.gov/resources/pue-comprehensive-examination-metric.

[18] A. I. Aravanis, A. Voulkidis, Metrics for assessing flexibility and sustainability of next generation data centers, in: 2015 IEEE Globecom Workshops (GC Wkshps), IEEE, 2015, pp. 1–6. URL: https://ieeexplore.ieee.org/document/7414182. doi:10.1109/GLOCOMW.2015.7414182.

[19] V. D. Reddy, B. Setz, M. Aiello, Metrics for sustainable data centers, IEEE Transactions on Sustainable Computing 2 (2017) 290–303. doi:10.1109/TSUSC.2017.2701883.

[20] T. van de Voort, V. Zavrel, J. I. Torrens Galdiz, J. L. M. Hensen, Analysis of performance metrics for data center efficiency: Should the power utilization effectiveness (pue) still be used as the main indicator? part 1, REHVA Journal 2017 (2017) 5–11. URL: https://www.rehva.eu/fileadmin/REHVA_Journal/REHVA_Journal_2017/RJ1/p.05/05-11_RJ1701_WEB.pdf.

[21] T. Brunschwiler, S. Paredes, B. Michel, Direct hot-water cooling for data centers, IBM Journal of Research and Development 55 (2011) 8:1–8:12. doi:10.1147/JRD.2011.2134079.

[22] P. Huang, et al., A review of data centers as prosumers in district energy systems: Renewable energy integration and waste heat reuse for district heating, Applied Energy 258 (2020) 114109. doi:10.1016/j.apenergy.2019.114109.

[23] Z. Liu, A. Wierman, Y. Chen, B. Razon, N. Chen, Data center demand response: Avoiding the coincident peak via workload shifting and local generation, Performance Evaluation 70 (2013) 770–791. doi:10.1016/j.peva.2013.08.014.

[24] T. Cioara, I. Anghel, M. Bertoncini, I. Salomie, D. Arnone, M. Mammina, T.-H. Velivassaki, M. Antal, Optimized flexibility management enacting data centres participation in smart demand response programs, Future Generation Computer Systems 78 (2018) 330–342. doi:10.1016/j.future.2016.05.010.

[25] C. Crozier, M. Liska, The potential of data center energy demand to provide grid flexibility, Current Sustainable/Renewable Energy Reports 12 (2025) 12. doi:10.1007/s40518-025-00258-9.

[26] Y. Zhang, et al., A survey of challenges and solutions for the integration of renewable energy in datacenters, Renewable and Sustainable Energy Reviews 155 (2022) 111787. doi:10.1016/j.rser.2021.111787.

[27] J. Jerez Monsalves, C. Bergaentzle, D. Keles, Impacts of flexible-cooling and waste-heat recovery from data centres on energy systems: A danish case study, Energy 281 (2023) 128112. doi:10.1016/j.energy.2023.128112.

[28] Y. Wang, et al., Data center integrated energy system for sustainability: Generalization, approaches, methods, techniques and future perspectives, The Innovation: Energy 1 (2024) 100014. doi:10.59717/j.xinn-energy.2024.100014.

[29] P. Terenius, P. Garraghan, R. Harper, A material social view on data center waste heat: Novel uses and metrics, Frontiers in Sustainability 3 (2023) 1008583. doi:10.3389/frsus.2022.1008583.

[30] X. Yuan, et al., Data center waste heat for district heating networks: A review, Renewable and Sustainable Energy Reviews 189 (2025) 113969. doi:10.1016/j.rser.2023.113969.

[31] Y. Hao, et al., Data centers waste heat recovery technologies: Review and evaluation, Applied Energy 384 (2025) 125489. doi:10.1016/j.apenergy.2025.125489.

[32] M. T. Takci, colleagues, Data centres as a source of flexibility for power systems, Electric Power Systems Research (2025). doi:10.1016/j.epsr.2024.110389.

[33] European Commission, Reporting requirements on the energy performance and sustainability of data centres, 2023. URL: https://energy.ec.europa.eu/publications/reporting-requirements-energy-performance-and-sustainability-data-centres-energy-efficiency_en, directive (EU) 2023/1791 and Delegated Regulation (EU) 2024/1364.

[34] The Green Grid Data Center Resource Effectiveness Standing Work Group, WP#93 Data Center Resource Effectiveness (DCRE) v1 Metric, Technical Report, The Green Grid, 2025. URL: https://www.thegreengrid.org/resources/library-and-tools/wp93-data-center-resource-effectiveness-dcre-metric, published 17 February 2025; accessed 2026-07-29.

[35] International Energy Agency, 2025, Energy and ai, URL: https://www.iea.org/reports/energy-and-ai.

[36] EPRI, Powering Intelligence: Analyzing Artificial Intelligence and Data Center Energy Consumption, Technical Report, Electric Power Research

Institute (EPRI), 2024. URL: https://www.epri.com/research/products/3002028905, published May 28, 2024.

[37] R. Ford, C. Maidment, C. Vigurs, M. J. Fell, M. Morris, Smart local energy systems (sles): A framework for exploring transition, context, and impacts, Technological Forecasting and Social Change 166 (2021) 120612. doi:10.1016/j.techfore.2021.120612.

[38] F. W. Geels, Micro-foundations of the multi-level perspective on socio-technical transitions: Developing a multi-dimensional model of agency through crossovers between social constructivism, evolutionary economics and neo-institutional theory, Technological Forecasting and Social Change 152 (2020) 119894. URL: https://www.sciencedirect.com/science/article/pii/S0040162518316111. doi:https://doi.org/10.1016/j.techfore.2019.119894.

[39] European Commission, 2019, Communication on the european green deal, URL: https://commission.europa.eu/publications/communication-european-green-deal_en.

[40] Interreg North Sea, 2025, Brave is building a €150m cleantech investment pipeline in seven regional energy systems, URL: https://www.interregnorthsea.eu/brave.

[41] Interreg North Sea Programme, Brave project, https://www.interregnorthsea.eu/brave, 2025. [Accessed 18-11-2025].

[42] M. Xu, et al., The impacts of eu ets on electricity system operation: Empirical evidence, Energy Policy 176 (2023) 113462. doi:10.1016/j.enpol.2023.113462.

[43] C. Koronen, M. Åhman, L. J. Nilsson, Data centres in future european energy systems—energy efficiency, integration and policy, Energy Efficiency 13 (2020) 129–144. doi:10.1007/s12053-019-09833-8.

[44] M. Vesterlund, et al., Data Center for Biomass Drying, Technical Report, RISE Research Institutes of Sweden, 2021. URL: https://urn.kb.se/resolve?urn=urn:nbn:se:ri:diva-54707, accessed: 2025-11-25.

[45] European Energy, 2025, Kassø e-methanol facility, URL: https://europeanenergy.com/kasso/.

[46] International Organization for Standardization, Iso/iec 30134 — information technology — data centres — key performance indicators, https://www.iso.org/standard/63498.html, 2016.

[47] S. Shao, J. Li, Y. Guo, X. Xu, A review of data center energy efficiency evaluation metrics, Energy and Buildings 266 (2022) 112111. doi:10.1016/j.enbuild.2022.112111.

[48] A. Safari, A. Abhari, A. Tizghadam, A systematic review of energy efficiency metrics for optimizing cloud data center operations and management, Electronics 14 (2025) 2214. doi:10.3390/electronics14112214.

[49] Uptime Institute, Uptime Institute Global Data Center Survey 2024, Technical Report, Uptime Institute, 2024. URL: https://datacenter.uptimeinstitute.com/rs/711-RIA-145/images/2024.GlobalDataCenterSurvey.Report.pdf, accessed: 2026-07-29.

[50] Smart Cities Marketplace, European Commission, 2024, Data centre supplies local heating – mäntsälä, finland, URL: https://smart-cities-marketplace.ec.europa.eu/insights/solutions/datacentre-supplies-local-heating-mantsala-finland.